From Text to Beta:
Macroeconomic Disclosures and Firm-level Systematic Risk

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June 2025

Acknowledgments: We thank Oliver Binz, Bill Cready, Ilia Dichev, Bjorn Jorgensen, Yaniv Konchitchki, Stan Markov, Suresh Nallareddy, Panos Patatoukas, Jedson Pinto, Peter Pope, Jason Schloetzer, Eddie Riedl, Kirti Sinha, Ayung Tseng, Paul Zarowin, Wuyang Zhao, and the workshop participants at The Accounting Design Project at Columbia University, the 44th Annual EAA Congress, FARS 2023, HARC 2023, the 2022 Lone Star Accounting Conference, Bocconi University, Emory University, Georgetown University, Michigan State University, Pennsylvania State University, University of Illinois at Chicago, University of California Berkeley, and University of Texas at Dallas for valuable comments and suggestions. An earlier version of this manuscript was titled: "When Micro Firms Speak Macro: Evidence on Firms' Macroeconomic Disclosures."

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Abstract

This study develops a firm-year measure of corporate performance sensitivity to future aggregate shocks using macroeconomic disclosures from 10-K filings. These disclosures (i) help quantify forward-looking earnings sensitivity to aggregate shocks more effectively than existing numerical and text-based proxies; (ii) reveal the economic drivers of such sensitivity ex ante, prior to realized macroeconomic changes; and (iii) capture priced systematic risk. Macroeconomic disclosures increase over time, vary with the business cycle, and exhibit significant within-firm variation—consistent with the time-varying nature of aggregate risk exposure. They are dispersed across multiple sections of the 10-K, underscoring the importance of contextualizing aggregate risk throughout the filing. Moreover, they help identify macro-relevant firms beyond existing proxies. Overall, corporate disclosures offer key insights into the economic drivers of firm-level systematic risk and cost of capital.

Key terms: Macroeconomy, Disclosure, Earnings Beta, Systematic Risk, Textual Analysis

Data sources: SEC filings, National Bureau of Economic Analysis, WSJ, Compustat, CRSP, I/B/E/S

I. INTRODUCTION

This study examines whether firms' macroeconomic disclosures can be used to quantify and explain—*ex ante*—earnings sensitivity to aggregate economic shocks and thus capture priced systematic risk.[1] We revisit central questions in accounting and finance. First, how sensitive are firms' earnings to future aggregate shocks? Second, what are the economic drivers of firms' sensitivity to such shocks? Third, is this risk priced in equity markets—i.e., does is command a risk premium? To answer these questions, we construct a novel firm-year measure—*MRisk*—based on macroeconomic disclosures in 10-K reports, that, *before* an aggregate shock results in macro changes: (i) helps quantify firms' earnings sensitivity to macro risk; (ii) identifies the economic source of such risk, and (iii) explains the cross-section of stock returns, thereby providing insights into firms' cost of capital.

Using corporate disclosures, this study takes a novel approach to a vast body of empirical research in accounting that has examined whether, to what extent, and how financial statement *numbers* convey information about firm-level systematic risk (e.g., Ball and Brown, 1967; Ball and Brown, 1968; Beaver et al., 1970; Ball et al., 2009; Penman and Zhang, 2020; Ellahie, 2021). While a growing literature has examined firm-level risk *disclosures* (e.g., Kravet and Muslu, 2013; Campbell et al., 2014; Hassan et al., 2019, 2023; Konchitchki and Xie, 2023), these studies have not focused on measuring and understanding priced systematic risk using macroeconomic discussions.

Systematic risk reflects the priced sensitivity—commonly referred to as "beta"—of an asset's payoffs to aggregate economic shocks. Given the well-documented limitations of "market beta" in explaining the cross-section of stock returns (e.g., Fama and French, 1992; Cochrane, 2011), studies in accounting have used reported numbers, including earnings and book value of equity, to assess priced systematic risk exposure. However, identifying ex ante the economic source of aggregate shocks driving systematic risk and measuring firms' priced sensitivities to those shocks remain persistent empirical challenges (e.g., Lewellen et al., 2010; Ball et al., 2021).

[1] Throughout the study, we (i) abbreviate "macroeconomic" as "macro," (ii) use "macroeconomic" and "aggregate" interchangeably, and (iii) refer to "earnings sensitivity to aggregate shocks," "macro sensitivity," and "earnings beta" as equivalent terms.

Even when existing measures of systematic risk appear to be priced, their underlying economic drivers often remain unclear. One of the most prominent examples of accounting-based characteristics used to proxy for systematic risk is the book-to-market ratio, which underpins Fama and French's (1993) widely used "High-Minus-Low" (HML) factor. While effective in explaining the cross-section of stock returns, the economic source of the risk captured by this factor is not well understood. In spite of ex-post rationalizations (e.g., Liew and Vassalou, 2000; Lettau and Ludvigson, 2001; Ball et al., 2020), there is no clear ex-ante rationale for why the book-to-market ratio should reflect sensitivity to any specific aggregate shock. More recently, Ellahie (2021) proposes priced measures of a firm's earnings sensitivity to aggregate earnings—commonly referred to as "earnings beta." [2] However, these measures similarly fall short of identifying, ex ante, the specific macro shocks driving aggregate earnings movements or explaining *why* firms display heterogeneous sensitivities to these shocks.

The challenge of identifying, measuring and understanding systematic risk arises from its time-varying nature. Both the economic sources of aggregate shocks and firms' sensitivities to those shocks evolve over time. For example, recessions can originate from fundamentally distinct causes—ranging from credit market disruptions to regulatory shifts and global health crises. Likewise, a firm's exposure to a given shock is itself time-varying, shaped by evolving strategic, technological, and operational decisions. Time-varying macro shocks and firms' time-varying exposures to such shocks make quantifying and understanding systematic risk ex ante very challenging (Ang and Chen, 2007). Thus, the literature has largely adopted an ex-post approach: researchers first identify firm characteristics that predict future returns and then retrospectively propose risk-based explanations for those return patterns (e.g., Hou et al., 2020; Bessembinder et al., 2025, Novy-Marx and Velikov, 2025).

Traditional approaches to identifying and measuring firm-level systematic risk face three key—not mutually exclusive—limitations: (i) they tend to be stable over time, (ii) they are inherently backward-looking, and (iii) they lack the ability to identify ex ante the specific nature of a firm's exposure to aggregate risk. The persistence of some firm characteristics, such as size or industry membership, is incompatible with inherently time-varying systematic risk exposure. Measures based on past

[2] See also Fama and French (1995) and Campbell and Vuolteenaho (2004).

covariances are similarly backward-looking and persistent. Moreover, firm-level covariances (using earnings), usually require 5 to 15 years of time-series data, which reduces sample size and introduces survivorship bias. Finally, reported numbers, covariances and many firm characteristics provide little or no ex-ante insight into the drivers of aggregate shocks and why certain firms are exposed to them more than others (i.e., why firms have different betas).

To address these limitations, our study leverages macroeconomic textual disclosures in annual reports to capture, ex ante, both the level and "nature"—i.e., the economic drivers—of firms' systematic risk embedded in earnings.[3] Our approach is an ex-ante one. That is, we use macro disclosures to first identify sources of aggregate risk and measure the firm's exposure to such risk. We then examine whether macro risk, as captured by corporate disclosures, indeed reflects earnings sensitivity to future macro shocks and priced systematic risk.

Specifically, we identify, each year, frequently used macroeconomic terms from an external sample of Wall Street Journal "Economic News" articles and count the occurrences of those terms in a sample of 10-K filings. We then scale the firm-level term counts by the total number of words in each filing. The resulting measure, *MRisk*, is designed to capture a time-varying and forward-looking earnings beta—i.e., a firm's earnings time-varying sensitivity to future aggregate shocks.

Examining systematic risk based on corporate disclosures offers several advantages over traditional approaches, both conceptual and empirical. Conceptually, our approach leverages management's private information about their firm's future macro exposure, incorporating expectations about upcoming strategic decisions. These disclosures allow to identify ex ante the economic drivers of systematic risk, such as exposure to credit markets or global trade shocks. Empirically, our approach does not rely on historical data and is therefore not subject to survivorship bias and it varies substantially by firm and year. Additionally, is it simple to replicate, transparent, and easily interpretable, requiring neither computationally intensive nor opaque techniques. Finally, it benefits from out-of-sample validation—drawing macroeconomic terminology each year from an external source unrelated to firms'

[3] From an economic standpoint, corporate disclosures are intended to provide context for and explain reported earnings. Accordingly, we assume that if managers possess information about their firms' systematic risk exposure, such information pertains primarily to the component of systematic risk embedded in accounting earnings (e.g., Ball and Brown, 1967; Ball and Brown, 1968; Ball et al., 2009).

own disclosures—while relying on commonly used terms and phrases to minimize the risk of omitting important macro words.

Our findings demonstrate that macroeconomic disclosures capture three key dimensions of systematic risk. First, they reflect firms' ex-ante performance sensitivity to macroeconomic shocks—thus providing a forward-looking earnings beta. Second, they reveal the economic drivers of upcoming macro shocks. Third, they contain information that explains the cross-section of stock returns, thereby providing insights that investors can utilize to estimate firms' cost of capital. To our knowledge, no existing approach in the literature successfully integrates all three elements to provide ex-ante insights into the types of macroeconomic shocks firms face each year, their heterogenous earnings sensitivity to such shocks, and demonstrate that these risks command positive risk premia—i.e., higher cost of capital.

We begin the empirical analyses by examining both aggregate and firm-level variation in macroeconomic disclosures. At the aggregate level, we find a rising trend in the relative proportion of macroeconomic discussions in 10-K filings, consistent with increasing firm–macro linkages and growing interconnectedness across firms over time (Acemoglu and Azar, 2020; Acemoglu et al., 2012). At the firm level, we observe substantial variation over time, across firms, and within firm over time. Notably, nearly half of the firm-level variation is time-varying, offering initial evidence of both shifting macroeconomic shocks and evolving firm sensitivities to those shocks.

Next, we show that managers discuss macroeconomic conditions across multiple sections of the 10-K filings. Notably, the "MD&A" and "Risk Factors" sections—the latter introduced only in 2005—account for just 50% of all macroeconomic disclosures. The remaining half appears in other sections, particularly "Business," "Footnotes," and "Market Risk." Each of these contributes roughly 15% of total macro-related content. This finding is significant, as most prior studies on risk disclosures have focused exclusively on the "Risk Factors," "MD&A," or both (e.g., Kravet and Muslu, 2013; Campbell et al., 2014; Konchitchki and Xie, 2023), overlooking a substantial portion of potentially relevant information.

A unique feature of macro disclosures is their potential to identify the economic drivers of a firm's exposure to systematic risk. Thus, we follow prior literature (Chen et al., 1986; Bonsall et al., 2013) to identify key macroeconomic factors—aggregate output, credit markets, prices, housing markets,

interest rates, and global trade—and categorize disclosures accordingly. We find that topic-specific macro disclosures vary over time in line with broad systematic trends. For example, credit-related disclosures start rising in 2007 thereby identifying ex ante the specific nature of the recession occurring during the 2008–2009 financial crisis. Notably, correlations across topics are low (i.e., 15%), suggesting they capture distinct macroeconomic shocks. Finally, average topic-level disclosure is strongly correlated with its cross-sectional dispersion, indicating that as macro risk increases, firms vary in the extent to which they disclose—consistent with heterogeneous risk exposure across firms and time.

We formally validate the ability of macro disclosures to capture, ex ante, future time-varying earnings sensitivities to aggregate shocks—i.e., whether they can serve as forward-looking earnings betas. Each year, we sort firms into quartiles based on lagged macro disclosure levels and regress portfolio-level earnings growth on contemporaneous macroeconomic indicators reflecting changes in nominal output (nominal GDP), real output (real GDP), and prices (inflation). Firms in higher disclosure quartiles exhibit stronger earnings sensitivity to macro conditions: both coefficient magnitudes and *R*-squared values increase monotonically across quartiles. For instance, the explanatory power of nominal GDP growth rises from 9% in the lowest disclosure quartile to 40% in the highest. These results are confirmed in firm-level regressions including stringent controls and fixed effects.

To demonstrate the relevance of our approach using macro disclosures, we benchmark it against 10 existing proxies for macro risk and/or earnings beta—four numerical and six text-based. The numerical proxies include: (i) firm size (e.g., Konchitchki and Patatoukas, 2014b); (ii) market beta (e.g., Fama and French, 1992); (iii) earnings beta (e.g., Ellahie, 2021); and (iv) a "bellwether" metric based on the correlation between firm earnings and a range of macroeconomic indicators (e.g., Hutton et al., 2012; Bonsall et al., 2013). The text-based proxies include risk disclosure measures based on word lists from (i) Campbell et al. (2014)—both the full list and a decomposition into systematic and idiosyncratic terms; (ii) Kravet and Muslu (2013); (iii) Loughran and McDonald (2011); and (iv) Hassan et al. (2019). Our measure provides incremental—and often stronger—information about firms' sensitivity to future macro fluctuations than alternative proxies, consistently across both portfolio- and firm-level analyses.

If macroeconomic disclosures help capture firms' future sensitivity to aggregate earnings, it is natural to expect that they may also identify macro-relevant or "bellwether" firms—those whose current

performance provides signals about future macroeconomic outcomes (e.g., Hutton et al., 2012; Bonsall et al., 2013; Konchitchki and Patatoukas, 2014b). We find evidence that *MRisk* identifies macro-relevant firms. Our analyses show that the accounting performance of firms with higher levels of macro disclosure correlates significantly more strongly with future changes in macroeconomic conditions.

Finally, we assess whether macro disclosures identify priced systematic risk. If a systematic risk is priced, it must be positively related to future returns and should exhibit a higher factor loading (beta) with a factor mimicking portfolio sorted based on risk exposure (Fama and French, 1993). We find that firms with higher levels of macro disclosure earn higher average returns in the subsequent year, indicating higher risk premia and therefore cost of capital. These firms also exhibit higher factor loadings with respect to a factor-mimicking portfolio constructed from macro disclosure. Therefore, our findings show that macro disclosures capture priced systematic risk.

This study makes several contributions to capital markets literature and the empirical asset pricing literature in accounting and finance. First, we show that macro disclosures provide unique ex-ante insights for understanding the economic drivers of and quantifying firms' earnings exposure to aggregate shocks—thus, systematic risk. Influential and recent contributions have relied on past correlations between firms' reported earnings and aggregate earnings (e.g., Ball and Brown, 1968; Beaver et al., 1970; Ball et al., 2009; Ellahie, 2021) or accounting-based characteristics, such as the book-to-market ratio (e.g., Chordia and Shivakumar, 2002; Penman and Zhang, 2020), to examine firms' systematic risk. However, these approaches cannot explain the economic drivers underlying systematic risk—*ex ante*, before aggregate shocks materialize (see Ball et al., 2021, p. 609). The literature largely makes indirect and ex-post rationalizations about the economic forces driving the non-diversifiable risk that existing measures capture (see Ellahie, 2021, p. 86). Thus, fundamental questions remain unanswered: Why do certain firms exhibit higher earnings betas? Why certain firm characteristics, such as the book-to-market ratio, are systematically related to future stock returns, and what underlying source of risk does it capture? Our findings show that firms' macro disclosures can answer these fundamental economic questions while providing a reliable measure of systematic risk that explains the cross section of stock returns. Overall, macroeconomic disclosures can help investors estimate and understand firms' cost of capital.

Second, this study contributes to the literature on textual risk disclosures (e.g., Kravet and Muslu, 2013; Campbell et al., 2014; Hassan et al., 2019; Konchitchki and Xie, 2023), while differing from prior work in several fundamental respects. Existing work has connected textual risk disclosures to a variety of future firm-level (idiosyncratic) risk outcomes (e.g., long-window returns volatility) or changes in investors' perceptions of risk (e.g., short-window market reactions). However, these studies do *not* examine whether or which risk disclosures relate to future returns and, therefore, to firms' systematic risk and cost of capital. At most, they document a correlation between disclosures and market beta (e.g., Campbell et al., 2014). Yet, as noted by Ellahie (2021, p. 85), market beta cannot explain variation in returns (see also Fama and MacBeth, 1973); thus, a positive relation between risk disclosures and market beta does not constitute a pricing test of disclosures about non-diversifiable risk.

Relative to prior risk disclosure studies, we also show that information relevant to systematic risk is disclosed throughout corporate filings rather than concentrated in designated sections (e.g., "Risk Factors" or "MD&A"). Therefore, accurately measuring firms' exposure to systematic risk requires a comprehensive analysis of the entire annual filing—an approach not typically adopted in prior studies (e.g., Kravet and Muslu, 2013; Campbell et al., 2014).

Third, we contribute to the growing literature on the macroeconomic information content of accounting numbers (e.g., Bonsall et al., 2013, Konchitchki and Patatoukas, 2014a; Kim et al., 2020; Hann et al., 2021; Abdalla and Carabias, 2022). This literature has primarily focused on identifying macro-relevant or "bellwether" firms and/or on how accounting numbers correlate with future economy-wide conditions. Our findings show that macro disclosures can be used to identify firms which forecasted earnings growth and special items better forecast future macro conditions. While prior studies have emphasized the predictive power of disclosure tone for macro outcomes (e.g., Jiang et al., 2019), we are the first to offer initial evidence on the unique role of macroeconomic disclosures in identifying firms' macro relevance over time.

II. BACKGROUNG AND MOTIVATION

Accounting Earnings and Systematic Risk

Systematic risk is a central topic of interest in modern portfolio theory and asset pricing due to its implications for expected returns, asset allocation, and risk management (e.g., Sharpe, 1964; Lintner, 1965; Merton, 1973; Fama and French, 1993, 1995, 1996). The financial economics literature has traditionally treated firm-level earnings news as primarily idiosyncratic and thus largely diversifiable, leading to the conclusion that aggregate earnings news plays only a marginal role for understanding and explaining systematic risk (e.g., Campbell, 1991; Vuolteenaho, 2002).

However, a prolific stream of accounting research—originating with Brown and Ball (1967; 1968), and further developed by Beaver et al. (1970), Ball et al. (2009), and more recently Penman and Zhang (2020), Ball et al. (2021), and Ellahie (2021)—has challenged this view on both conceptual and empirical grounds. Conceptually, this literature argues that because corporate profits represent a significant share of changes in aggregate output, it is reasonable to expect that changes in earnings may contribute to our understanding of aggregate—and therefore non-diversifiable—risk. Empirically, these studies show that there is substantial systematic variation embedded in firm-level earnings changes and that changes in earnings are closely linked to changes in prices—that is, expected returns. Taken together, this body of research highlights how variation in firm-level earnings can be used to understand firm-level exposure to systematic risk.

While these studies document the presence of systematic risk in earnings, accurately measuring a firm's sensitivity to such risk, and understanding the underlying sources, remains an open challenge. As Ball et al. (2021, p. 609) note, while prior research in accounting has identified earnings-based measures of systematic risk that help explain the cross-section of stock returns, these studies have had limited success in identifying *ex ante* the economic forces underlying systematic risk in earnings. This limitation stems from the fact that both the aggregate shocks underlying systematic risk and firms' responses to those shocks are time-varying, making it difficult to capture firms' future sensitivity to economic dynamics using inherently backward-looking measures such as accounting earnings.

Limitations and Economic Interpretability of Conventional Measurement Approaches

Measuring and understanding systematic risk, including risk in earnings, is inherently challenging due to the time variation in both the economic forces underlying macroeconomic shocks and firms' responses to those shocks. Firms are subject to different economic shocks over time. Consider recent economic cycles: the late 1990s witnessed the boom and subsequent bust of internet-based firms; the 2000s were characterized by a surge in real estate prices that ultimately led to a credit crisis; and more recently, the global economy was disrupted by the COVID-19 pandemic. While these episodes may exhibit comparable effects on aggregate output, the underlying shocks differ fundamentally, leading to heterogeneous firm-level exposures. Furthermore, even when the nature of the shock remains constant, firm responses may vary substantially. For instance, a company can alter its sensitivity to oil price fluctuations by transitioning its fleet to electric vehicles. In sum, firms' sensitivity to aggregate shocks is highly time-varying, it is firm-year specific, and therefore difficult to measure and explain ex ante, that is before an aggregate shock occurs that impact the firm.

Traditionally, accounting and finance studies have measured systematic risk using two approaches or a combination of these approaches (Ball et al., 2021). The first consists in using past time-series covariances between firm-level earnings and aggregate earnings or other macroeconomic state variables (e.g., Ellahie, 2021). The second approach is based on using portfolios sorted on variables that are known to predict returns and are thus likely candidates to capture some latent systematic risk (e.g., Penman and Zhang, 2020). While informative, these approaches face key limitations.

First, measures based on past covariances are inherently backward-looking. They rely on historical time-series correlations and on earnings figures that themselves reflect past performance. As a result, they provide an incomplete characterization of firms' time-varying sensitivity to *future*—and evolving—macroeconomic shocks. Second, these estimates typically require a minimum of five, and often ten to fifteen, time-series observations to generate reliable firm-level measures of sensitivity to aggregate conditions. This requirement restricts the sample, limits the generalizability of empirical findings, and introduces survivorship bias into the analysis. Moreover, depending on the number of observations used for estimation, such proxies tend to be relatively static and therefore poorly suited to capturing the inherently time-varying nature of aggregate shocks and firms' evolving responses to them.

Third, while using portfolios to estimate covariances may address concerns related to sample size and survivorship bias, both firm-level and portfolio-based covariance approaches fail to shed light on the underlying economic drivers of firms' exposure to systematic risk. As a result, these measures lack *economic interpretability* and are likely to produce noisy signals of future systematic risk exposure (Ogneva, 2013, p. 268).

Notably, the lack of economic interpretability of systematic risk measures that explain the cross-section of stock returns also represents a key limitation in the empirical asset pricing literature (e.g., Fama and French, 1993). Most studies adopt an ex-post approach to identifying and understanding systematic risk, often relying on firm characteristics that serve as proxies for latent or unidentified risk factors. For example, the "High-Minus-Low" (HML) factor developed by Fama and French (1993) builds on Basu (1977), who finds that the book-to-market ratio predicts returns. Fama and French (1992, 1993) construct book-to-market–sorted portfolios to explain return variation across firms. However, the economic source of risk captured by HML remains unclear. Moreover, the literature disagrees on whether HML reflects compensation for risk or rather mispricing driven by behavioral biases (e.g., Lakonishok et al., 1994). In response, subsequent research has attempted to link known pricing anomalies—such as the value premium—to macroeconomic risk proxies, including business cycle indicators (Liew and Vassalou, 2000; Chordia and Shivakumar, 2002) and the consumption-to-wealth ratio (Lettau and Ludvigson, 2001). Yet, these efforts are largely *ex-post* rationalizations: there is no clear *ex-ante* rationale for why the book-to-market ratio should proxy for sensitivity to any specific aggregate shock.

This ex-post approach to explaining why certain risk factors appear to be priced has contributed to the proliferation of return-predictive variables—what Cochrane (2011) and others have referred to as the "factor zoo" (see also Hou et al., 2020). Such proliferation reflects a disconnect between theoretical asset pricing foundations and empirical practice. The ex-post approach may also help explain why many of these identified characteristics lose their predictive power over time (Chordia et al., 2014; Engelberg et al., 2018). In contrast to the ex-post approach, our approach first identifies the nature of the macro risk as well as firms that are more exposed to the risk. We then examine whether these risks are reflected

in a higher earnings sensitivity to macroeconomic changes as well as higher cost of capital. That is, we first identify aggregate risks and then examine whether the risks are priced.

Prior Research on Textual Risk Disclosures

Accounting earnings are typically accompanied by managerial disclosures that contextualize and explain those figures—including press releases, annual and quarterly filings, and earnings conference calls. It is therefore reasonable to expect that a nontrivial amount of information about firms' systematic risk may be conveyed through the textual disclosures that accompany financial statement numbers. The key advantage of such disclosures is that, unlike accounting earnings—which are inherently backward-looking—they are forward-looking to varying degrees (e.g., Li et al., 2013; Campbell et al., 2014; Bushman et al., 2015). As a result, they can offer valuable ex-ante insight into both firms' future exposure to systematic risk and the underlying economic nature of that exposure—an area of inquiry that the existing literature has found particularly challenging to tackle.

Prior research has examined risk factor disclosures extensively—both within 10-K filings and across various disclosure channels such as earnings conference calls (e.g., Kravet and Muslu, 2013; Campbell et al., 2014; Hope et al., 2016; Hassan et al., 2019; 2023; Konchitchki and Xie, 2023; Lyle et al., 2023). However, this literature has not focused on measuring or understanding systematic risk, and therefore has not identified and explained information investors can use to estimate a firm's cost of capital. Several studies have linked firm-level disclosures to changes in investors' perceptions about risk—without explicitly distinguishing between systematic and idiosyncratic components—by examining the relationship between risk factor disclosures and short-window market reactions (e.g., Kravet and Muslu, 2013). In addition, this stream of research has explored the determinants of risk factor disclosures and connected such textual information to future firm-level outcomes, such as idiosyncratic volatility (e.g., Campbell et al., 2014).

Notably, some studies have also attempted to isolate disclosures more closely related to systematic risk. For instance, Campbell et al. (2014) show that words manually classified as pertaining to systematic risk within the "Risk factors" and "MD&A" sections of 10-K filings are associated with future market beta. However, these studies do not directly examine whether the risks captured by these

subsets of words are priced—i.e., whether they are related to future stock returns. In fact, despite its theoretical appeal, market beta is well known to perform poorly in explaining the cross-section of stock returns, and thus fails to capture information that investors may find useful in assessing a firm's systematic risk exposure. As Ellahie (2021, p. 85) aptly notes, a positive association between risk disclosures and market beta does not constitute a pricing test of disclosures related to non-diversifiable risk. In contrast to prior research on risk disclosures, our study focuses, specifically, on textual information that can inform investors about firms' cost of capital.

It is also worth noting additional limitations of prior research on risk disclosures within the context of systematic risk analysis, which our measurement approach addresses, as detailed in the following section. One limitation is that the characterization of words pertaining to systematic risk has typically not been validated using an external, time-varying reference sample—such as economic news articles on the Wall Street Journal—that could capture evolving aggregate shocks. In addition, prior studies have generally focused on specific sections of the 10-K report, such as "Risk Factors" and "MD&A," which may result in omitting a substantial portion of information relevant to systematic risk.

More recent studies have examined specific dimensions of aggregate risk within the context of firm-level risk disclosures. For instance, Hassan et al. (2019; 2023) and Konchitchki and Xie (2023) investigate political, pandemic, and inflation risk, respectively. In parallel, recent finance research has analyzed the "Risk Factors" section of annual filings using unsupervised machine learning techniques to identify combinations of idiosyncratic and systematic risks and assess their relevance for asset pricing (Lopez-Lira, 2023). While informative, these studies do not focus on firms' earnings sensitivity to the macroeconomy, nor on the asset pricing implications of macroeconomic disclosures for systematic risk.

Overall, while prior research generally concludes that risk factor disclosures are informative to various degrees, no prior study directly links macro discussions to firms' earnings sensitivity to aggregate economic shocks as a means to explain and quantify firm-level systematic risk and cost of capital.

The Role of Macroeconomic Disclosures

Among textual disclosure channels, we choose to study macro discussions within 10-K regulatory filings because of their relevance and availability. To begin with, 10-K reports represent the most comprehensive source of companies' annual disclosures. If macroeconomic conditions affect the firm's performance, and managers provide private information regarding their firm's exposure to such conditions, we expect the disclosure to appear within annual filings. Also, because these are mandated filings, they are available for a large cross-section of U.S. firms over time and are less subject to self-selection issues compared to alternative disclosures (e.g., conference calls).

While prior studies generally agree that risk factor disclosures contain substantial information content, no study has shown that financial disclosures can be used to interpret ex ante firm-level *priced* sensitivity to aggregate shocks—that is, systematic risk. In fact, both the accounting and empirical asset pricing literatures have long grappled with the challenge of understanding ex-ante firm-level exposure to systematic risk.

There are at least three reasons to question whether corporate disclosures can meaningfully reflect time variation in firm-level sensitivity to macro shocks. First, managers are not professional macroeconomic forecasters. Information intermediaries such as analysts are often presumed to possess superior macroeconomic insights (e.g., Darrough and Russell, 2002; Piotroski and Roulstone, 2004; Kadan et al., 2012; Hutton et al., 2012; Ali et al., 2023). Similarly, Konchitchki and Xie (2023) document that firms often fail to fully disclose their sensitivity to inflation shocks.

Second, managerial disclosures may be strategic in nature. Prior research shows that managers are more likely to discuss adverse economic events when they can attribute poor performance to external factors (Schloetzer et al., 2021). Moreover, disclosures are often criticized as being repetitive or boilerplate (Brown and Tucker, 2011), limiting their usefulness in capturing meaningful variation in exposure to macroeconomic shocks. This issue is particularly important in the context of time varying risks. If the disclosures are indeed boilerplate or sticky, they will fail to capture the time varying component of risk, rendering disclosures uninformative.

Third, even if managers can identify and disclose their firms' macroeconomic risk exposures, it remains an open empirical question whether these exposures are *priced*—that is, whether they are

associated with future stock returns. Not all exposures to macroeconomic shocks are necessarily priced, only the ones that investors demand a risk premium for. Thus, testing the relation between disclosures and the cost of capital is a critical step, with unknown a priori results.

In sum, whether managers possess decision-useful information about their firms' time-varying exposure to systematic risk—and whether they are willing and able to communicate this information in annual filings—is an empirical question. Our study addresses this gap by examining whether macroeconomic disclosures in annual reports convey information about priced earnings sensitivity to aggregate shocks.

III. DATA AND METHODOLOGY

We use three primary data sources. Textual data is extracted from (i) 10-K filings obtained from the *SEC EDGAR* database, and (ii) *Wall Street Journal* (WSJ) "Economic News" articles downloaded from the *Dow Jones Factiva* repository. Firms' financials are primarily from *WRDS Compustat* (annual files) and *CRSP* (daily files). Macroeconomic data is primarily obtained from the *Bureau of Economic Analysis* at the U.S. Department of Commerce.

We measure macro disclosure using textual analysis of a firm's 10-K filing. First, we download 10-K reports filed between 1994 and 2021 with valid CIK, GVKEY, and PERMNO identifiers and a minimum length of 500 words. We exclude 10-K Amendments, 10-KSB, and 10-KSB Amendments. We also exclude all financial firms (i.e., SIC codes 6000-6999) due to their unique business model and reporting rules. Finally, we drop observations with missing Compustat or CRSP items and lagged disclosures (i.e., a 10-K report in year *t-1*). We obtain a large sample of 86,624 firm-year observations with valid disclosure and financial data; the sample selection details are reported in Table 1.

To create the macroeconomic disclosure measure, we compute the number of occurrences of one and two-word phrases (i.e., "unigrams" and "bigrams") that depict macroeconomic conditions within 10-K filings. Notably, we scale the total count of macro terms by the total number of words within the report to control for the general propensity to disclose within a 10-K filing. Our approach is similar, for example, to Li et al. (2013) and Bushman et al. (2015) that use textual analysis to measure a firm's

competitive environment by counting the number of competition-related terms in firms' 10-K reports. Thus, the resulting text-based measure of disclosed macroeconomic conditions is:

$$MRisk_{i,t} = \frac{MacroWords_{i,t}}{10kWords_{i,t}} \; x \; 1{,}000$$

where *MacroWords* and *10kWords* are the total number of macroeconomic words and the total number of words within firm's *i* 10-K for year *t*, respectively. We multiply the ratio by 1,000 to enhance readability.

The intuition underlying *MRisk* is straightforward. We hypothesize that when a manager expects her firm's future performance to exhibit higher sensitivity to macroeconomic conditions, she will offer more extensive macroeconomic discussions in a given year regarding the specific macroeconomic conditions the firm is exposed to. In other words, we conjecture that firms' macro disclosures contain unique forward-looking information regarding a firm's earnings time-varying exposure to macroeconomic shocks and thus, these disclosures can inform about priced systematic risk.

We identify macroeconomic words in three steps. First, we select the top 1% macroeconomic terms—by yearly frequency rank—within a large sample of 22,993 *Wall Street Journal* "Economic News" articles published between 1994 and 2021. We exploit the *Dow Jones Factiva* categorization of WSJ articles and focus our data collection on "Economic News", a broad category including articles on the general macroeconomic environment, monetary policy, and domestic or international trade. We exclude short newspaper articles: those characterized by less than 5 sentences and 100 words. In particular, we focus on one and two-word phrases (i.e., "unigrams" and "bigrams") excluding "stop-words" (e.g., prepositions, conjunctions); see Frankel et al. (2017). Next, we extract the top 1% most frequent one and two-word phrases within the sample of 86,624 10-K reports. Finally, we select the intersection of terms found in *both* the WSJ and 10-K samples and make those terms our target macroeconomic words. Appendix B includes the word list, while Appendix C reports three examples of 10-K macro disclosures.[4]

[4] One could design a more accurate yet more complex procedure. For instance, relatively infrequent macroeconomic words could be informative about the macro sensitivity of the firm. To include the less frequent words, natural language processing research suggests applying weighting schemes such as TF-IDF (i.e., "Term

We highlight three key measurement choices that differentiate our empirical approach from prior research on risk disclosures. First, we rely on a large sample of *Wall Street Journal* "Economic News" articles to identify, for each year in our sample, terminology that is *a priori* associated with the macroeconomy. This allows us to validate the set of macroeconomic terms using a source external to the firms' disclosures in our sample. This approach contrasts sharply with that of influential studies on risk factor disclosures for two reasons. First, it does not rely on the disclosures themselves—which are subject to managerial discretion—to identify systemically relevant terminology. Second, the set of macroeconomic terms varies year by year, capturing the time-varying nature of aggregate shocks. In contrast, for example, Campbell et al. (2014) use a clustering algorithm to identify words frequently used by managers in risk disclosures across the entire pooled sample and then manually classify them as relatively more or less idiosyncratic, without accounting for temporal variation.

Second, we count macro terms across all sections of the 10-K, whereas most prior studies focus on a limited subset of sections. For instance, Kravet and Muslu (2013), Campbell et al. (2014), Konchitchki and Xie (2023), and Lyle et al. (2023) focus on one or more sections—most commonly "Risk Factors" and "MD&A"—to study idiosyncratic or relatively more systematic disclosures. Our inclusive approach reflects the fact that managers often provide contextually related macroeconomic content throughout the filing.

Third, we focus specifically on disclosures related to macroeconomic risk exposure, rather than broader risk-related language (e.g., Kravet and Muslu, 2013; Campbell et al., 2014) or disclosures tied to particular aggregate topics (e.g., Hassan et al., 2019; 2023; Konchitchki and Xie, 2023). Our focus aligns with the objective of developing a measure that more directly captures earnings sensitivity to aggregate shocks, and is therefore better suited to identifying priced systematic risk.

Frequency-Inverse Document Frequency." See also Hassan, et al., 2019). We believe that increased sophistication and complexity comes at the cost of replicability and interpretability, which we consider key features of empirical research (e.g., Hail, Lang, and Leuz, 2020). We therefore opt for a reliable, objective, and easily interpretable procedure, and consider our results a lower bound for the role of annual reports in capturing firms' earnings sensitivity to aggregate shocks.

IV. DESCRIPTIVE ANALYSIS

Sample Statistics

Table 1, Panel B presents descriptive statistics. *MRisk* has a mean (median) of 1.90 (1.64) with a standard deviation of 1.22. Firms use an average (median) of 61 (45) macroeconomic words and 30,837 (28,400) total words per filing. For comparison, we calculate risk factor disclosures (RFD_{CEA}) using a similar scaling approach—counting Campbell et al. (2014) risk words per 1,000 words in the 10-K. The mean RFD_{CEA} is 16.38, much higher than *MRisk* (1.90), highlighting that macro disclosures capture a distinct, likely less idiosyncratic dimension of firm risk.

The median firm in our sample has a market capitalization of $350 million and *MRisk* is positively associated with other proxies for systematic risk: raw (ranked) correlations are 0.32 (0.32) with firm size, 0.14 (0.15) with market beta, and 0.14 (0.25) with earnings beta. *MRisk* is also positively correlated with risk factor disclosures at 0.18 (0.25) and negatively with idiosyncratic volatility at –0.20 (–0.23). For brevity, these results are not reported.

Aggregate Time Trend

We use *MRisk* to examine the aggregate time trend in macroeconomic disclosures within 10-K filings. Figure 1, Panel A shows a clear upward trend, with notable spikes during recessions—around the 2001 tech downturn, the 2007–2009 financial crisis, and the 2020 COVID-19 pandemic. This pattern aligns with theoretical and empirical work on increasing firm-macro interlinkages over time (Acemoglu et al., 2012; Acemoglu and Azar, 2020). The rise in disclosures during major economic shocks suggests that firms respond to heightened aggregate uncertainty by providing more macro-related information. Overall, the cyclical variation in *MRisk* supports its potential validity as a proxy to measure and understand firms' earnings sensitivity to the macroeconomy and, therefore, systematic risk exposure.

Notably, our primary goal, in examining macro disclosures, is to quantify and better understand firms' time-varying earnings beta—that is, their earnings sensitivity to future aggregate shocks—rather than to examine year-over-year changes in beta. This objective motivates our empirical focus on

analyzing the level of macro disclosures, both graphically and statistically, rather than changes in disclosure levels.

Location of Macroeconomic Disclosures

We next examine where macroeconomic disclosures appear within the 10-K to assess whether our decision to analyze the entire filing—rather than focusing solely on the "Risk Factors" or "MD&A" sections, as is common in the literature (e.g., Campbell et al., 2014; Konchitchki and Xie, 2023)—was appropriate. Using HTML formatting and SEC-defined tags, we divide each 10-K into 20 standardized sections (from Item 1, Business, to Item 15, Exhibits). We scrape each document using start and end signals (e.g., "Item 7A" to "Item 8") to isolate section boundaries and count macroeconomic terms in each. This allows us to determine whether macro-related content is concentrated in specific sections or dispersed throughout the filing—thus validating or challenging our methodological choice.[5]

Figure 2, Panel A shows the average relative frequency of macroeconomic disclosures across 10-K sections. We find that macro terms appear throughout the filing, with the "Business," "Footnotes," and "Market Risk" sections accounting for nearly 50% of all macro content. In contrast, the "Risk Factors" (introduced in 2005) and "MD&A" sections contain less than half of macro-relevant terms. While the share of macro content in "Risk Factors" increases over time—partly reflecting a shift from the "Business" section—our findings indicate that managers discuss macroeconomic exposure across multiple sections. This widespread distribution suggests that limiting analysis to "Risk Factors" or "MD&A," as prior literature often does, may omit relevant information about systematic risk exposure. Overall, our results underscore the importance of analyzing the full 10-K to comprehensively capture firm-level exposure to macroeconomic shocks.

Firm-level Variation

Firm-level sensitivity to aggregate shocks—and thus systematic risk—is known to vary over time (e.g., Ang and Chen, 2007; Ball et al., 2009; 2021).[6] If managers disclose macroeconomic conditions

[5] This analysis extends prior research examining disclosure topics within 10-K reports (e.g., Dyer et al., 2017).

[6] A notable example is the "Book-to-Market Effect" (Fama and French, 1996): although betas are estimated over the full sample, firms regularly move across book-to-market portfolios, implying time-varying sensitivity to the underlying risk factor.

based on private insights into their firm's exposure to such shocks, we should observe meaningful firm-level variation in *MRisk* and its components. To test this, we assess whether *MRisk* displays within-firm variation, a necessary feature for any measure aiming to capture priced, time-varying systematic risk.

We begin with a variance decomposition. Table 2, Panel A shows that approximately 70% of *MRisk* variation occurs at the firm level, unexplained by time, industry, or time × industry fixed effects.[7] Importantly, about 40% of this firm-level variation is time-varying—highlighting a desirable property for a measure of time-varying firm-level exposure to aggregate shocks.

We complement this analysis with a transition matrix. Each year, firms are sorted into quartiles based on their *MRisk* score, and we examine their likelihood of remaining in the same quartile the following year (Table 2, Panel B). We find substantial mobility: firms in the second and third quartiles have a 40% probability of switching quartiles, while even those in the first and fourth quartiles show a 25% chance of moving. These patterns reinforce two key points: (i) *MRisk* exhibits notable within-firm variation over time, and (ii) even firms with seemingly low or high macro exposure experience shifts in sensitivity, underscoring the importance of a time-varying measure to capture aggregate risk.

V. HOW SENSITIVE ARE FIRMS' EARNINGS TO FUTURE AGGREGATE SHOCKS?

Portfolio-level Analyses

Having established several desirable properties of macro disclosures, we next examine whether they capture firms' earnings sensitivity to macroeconomic conditions—a necessary condition for reflecting aggregate risk. For each year t, we sort firms into quartiles based on lagged *MRisk* ($t-1$) and test whether changes in aggregate conditions relate to changes in accounting earnings. This non-parametric, portfolio-based approach—common in empirical asset pricing (e.g., Fama and French, 1993; Cochrane, 2011)—accommodates non-linearities and improves interpretability. We use quartiles to form relatively large portfolios that better reflect the systematic component of earnings, which are more skewed than returns due to infrequent but extreme charges (Ball et al., 2009; 2021). Using finer partitions increases measurement error, particularly for low macro-disclosure firms, where small word count differences

[7] We use detailed industry definitions, up to 4-digit SIC, covering about 400 sectors.

can result in misclassification. We expect firms with higher *MRisk* to show greater earnings sensitivity to macro conditions, consistent with managerial awareness of aggregate risk exposure.

Specifically, we sort firm-year observations into quartiles of lagged *MRisk* and regress value-weighted changes in operating income on contemporaneous measures of (i) nominal GDP growth, (ii) real GDP growth, and (iii) inflation.[8] We estimate the following model:

$$\Delta OI_{p,t} = \beta_0 + \beta_1 \, \Delta \, Macro_t + \varepsilon_t \qquad (1)$$

$\Delta OI_{p,t}$ is the value-weighted portfolio average change in firm-level operating income in year *t*, scaled by beginning-of-year stock price. *Δ Macro*, is alternately measured as: (i) nominal GDP growth, (ii) real GDP growth, or (iii) inflation growth. Each time-series regression includes 27 annual observations.[9]

Table 3 reports the results. For firms in the highest *MRisk* quartile, changes in operating income are positively and significantly associated with all three macro indicators. More importantly, we observe a monotonic increase in both coefficients and *R*-squared across *MRisk* quartiles. For instance, the coefficient on real GDP rises from 0.03 in the lowest quartile to 0.13 in the highest, while *R*-squared increases from 0.09 to 0.32. Similar patterns hold for nominal GDP and inflation, and tests confirm that the differences between quartiles 1 and 4 are statistically significant. These findings suggest that *MRisk* effectively captures firms' sensitivity to future macroeconomic conditions.

Alternative Proxies

We next assess *MRisk* ability to identify firms with higher future macro sensitivity—i.e., firm-level earnings sensitivity to aggregate shocks—relative to existing measures. We benchmark *MRisk* against ten alternatives: four numerical and six textual proxies. The numerical measures include: (i) *Size* (e.g., Konchitchki and Patatoukas, 2014b), (ii) *Market Beta* (e.g., Fama and French, 1992), capturing exposure to market returns, (iii) *Earnings Beta*, reflecting firms' past sensitivity to aggregate earnings

[8] We find similar results with changes in industrial production and unemployment.

[9] Our choice to use operating income is intended to capture the most persistent component of earnings, aligning with prior literature (e.g., Ball et al., 2021).

(Ellahie, 2021),[10] and (iv) *Bell*, which measures the time-series correlation between firm earnings and macroeconomic variables such as GDP (e.g., Hutton et al., 2012; Bonsall et al., 2013).[11]

The textual proxies include: (i) RFD_{CEA}, the total Campbell et al. (2014) risk words from "Risk Factors" and "MD&A;" (ii) $RFD_{CEA\ SYST}$, the subset identified as systematic; (iii) $RFD_{CEA\ IDIO}$, the subset identified as idiosyncratic; (iv) RFD_{KM}, Kravet and Muslu (2013) risk words from "Risk Factors;" (v) RFD_{LM}, Loughran and McDonald (2011) uncertainty terms within the 10-K; and (vi) *PRD*, the top-120 political risk bigrams found in annual filings from Hassan et al. (2019). All counts are scaled per 1,000 words in each firm's 10-K.

Table 3 shows that *MRisk* offers more precise information relative to alternative proxies, both numbers- and text-based. In fact, the univariate portfolio analyses show that *MRisk* exhibits: (i) a monotonic increase in coefficients and *R*-squared across quartiles, and (ii) a larger spread in explanatory power between the highest and lowest quartiles. Monotonicity is key.[12] A reliable proxy for macro sensitivity should consistently rank firms along the intensity of their exposure. When the relationship between the proxy and macro sensitivity is non-monotonic, it introduces noise and misclassification—undermining the interpretation of differences across quartiles. For example, *Size* shows some predictive power only in the top quartile, offering little distinction across the remaining 75% of firms. This results in misleading group assignments and diminishes the value of statistical comparisons across quartiles.

We emphasize that, due to the non-monotonic pattern in coefficients and *R*-squared observed for the alternative proxies, we do *not* conduct a two-way conditional sort. Such an approach would require ranking firms first on a noisy signal, then on *MRisk*, resulting in a noisy and potentially misleading classification. Instead, to assess the incremental information content of *MRisk* with regard to firms' macro sensitivity, we conduct firm-level regressions with saturated controls and fixed effects, as described in the following paragraph.

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[10] Our results do not change when employing the array of earnings beta measures examined by Ellahie (2021).

[11] Following Bonsall et al. (2013), we estimate firms' earnings sensitivity to macro conditions and rank firms into quartiles based on the regression R^2—defining the *Bell* measure from 1 (low sensitivity) to 4 (high sensitivity). We use the same macro variables as in their study, including interest rates, inflation, GDP, unemployment, housing starts, and industrial production.

[12] We note that this evaluation approach is quite common within the finance literature (e.g., Fama and French, 1996). Several influential works, in fact, examine the statistical difference between "high" and "low" characteristic-based portfolio returns after assessing a monotonic trend across the relevant portfolios.

Firm-level Analyses and Incremental Information Content

We design OLS regressions that allow to examine if *MRisk* offers incremental information content regarding firms' performance sensitivity to aggregate shocks controlling for: (i) time-varying fundamentals, (ii) time-varying industry shocks, (iii) time-invariant firm characteristics; (iv) existing numerical proxies of sensitivity to aggregate shocks, and (v) alternative text-based proxies. We estimate the following pooled model:

$$\Delta OI_{i,t} = \beta_0 + \beta_1 \Delta Macro_t + \beta_2 MRisk_{i,t-1} + \beta_3 \Delta Macro_t \times MRisk_{i,t-1} + \sum_{j=4}^{J} \beta_j Alternative\ Proxies_{i,t-1} + \sum_{k=J+1}^{K} \beta_k \Delta Macro_t \times Alternative\ Proxies_{i,t-1} + \sum_{v=K+1}^{V} \beta_v Controls_{i,t-1} + \sum_{q=V+1}^{Q} \beta_q \Delta Macro_t \times Controls_{i,t-1} + \sum_{n=Q+1}^{N} \beta_n Controls_{i,t} + Firm\ FE_i + Year \times Sector\ FE_{s,t} + \varepsilon_{i,t} \quad (2)$$

where $\Delta Macro$ represents, alternatively, growth in GDP, nominal and real and inflation, $Alternative\ Proxies$ include all of the ten alternative variables previously employed to capture macro sensitivity, and $Controls$ are meant to capture firm characteristics likely associated with earnings' sensitivity to aggregate shocks, including book-to-market, ROA, leverage, special items, number of geographic segments, and idiosyncratic stock returns volatility over the prior 12 months.[13]

The main coefficient of interest is β_3, which captures whether the sensitivity of firm-level earnings changes to GDP growth varies with the relative proportion of macroeconomic disclosures in 10-K reports. This interaction term reflects the incremental information content—i.e., macro sensitivity—conveyed by macroeconomic disclosures, conditional on alternative proxies and control variables. Importantly, all alternative proxies and controls are also interacted with GDP growth, and all variables are standardized to enhance comparability and facilitate interpretation of the economic significance of the estimated coefficients. Furthermore, the inclusion of firm fixed effects and industry-by-year fixed effects accounts for time-invariant firm characteristics and time-varying industry-specific trends that

[13] In sensitivity analyses we also employ ranked (in quartile portfolios) variables. We do so because raw variables can be both positive and negative. This generates uncertainty in the interpretation of the interaction terms which would be positive if both variables are positive, but also if both are negative. Our results remain consistent.

may influence macro sensitivity.[14] Therefore, the effect we estimate is driven by non-persistent, firm-level variation in *MRisk*.

The results in Table 4 indicate that *MRisk* provides incremental explanatory power for firms' performance sensitivity to macroeconomic shocks, even after controlling for a broad set of firm characteristics and alternative sensitivity measures. The interaction between *MRisk* and GDP growth is consistently positive and statistically significant across all model specifications. These findings confirm the effectiveness of *MRisk* in firm-level regressions. Notably, no other proxy captures earnings sensitivity to aggregate shocks as convincingly—both in terms of economic magnitude and statistical significance.

Additional Analyses and Sensitivity Tests

We conduct a range of additional analyses to assess the robustness and interpretability of our findings. First, we examine the temporal stability of *MRisk* by sorting firms using values from years t, $t-1$, and $t-2$ and re-estimating our main models. We find that the monotonic trend in explanatory power and incremental information content largely disappear after two years, suggesting that firms' macro sensitivity is time-varying, supporting the use of a firm-year-specific measure. Second, we test whether *MRisk* predicts future macroeconomic conditions, as opposed to capturing sensitivity to them, and find no such predictive power, in line with prior work (e.g., Jiang et al., 2019). This confirms that *MRisk* reflects perceived sensitivity to macroeconomic conditions rather than managerial forecasts of aggregate changes. Third, we assess the contribution of specific 10-K sections by excluding macro terms from Item 1A ("Risk Factors") and by isolating macro disclosures from individual sections such as the "MD&A" and "Business" description. While excluding "Risk Factors" marginally weakens results, the main findings persist. When macro disclosures are derived from individual sections *only*, results lose significance, highlighting the importance of aggregating macro content across the full 10-K, as managers often provide contextual information throughout the filing. Fourth, we expand our macro term list to include slightly less frequent but relevant terms (e.g., "labor market,"

[14] We note that our firm-level OLS results are unchanged when we (i) only include firm fixed effects, and (ii) interact firm (or industry) fixed effects with $\Delta\, Macro$ thus making the model fully interacted.

"unemployment") and find consistent results, with monotonic increases in coefficients and explanatory power across *MRisk* quartiles in portfolio-level analyses and incremental information content in firm-level regressions. Fifth, given that the largest component of *MRisk* reflects the total number of words in the 10-K, we examine whether our results are driven by overall document length rather than the relative frequency of macroeconomic terms. Robustness tests confirm that the findings are indeed driven by macro-related content, not by total word count.

Finally, to further illustrate the economic relevance of *MRisk*, we test its macroeconomic information content—specifically, its ability to identify "bellwether" firms, i.e., firms whose performance helps forecast future macroeconomic conditions. We focus on two firm-level fundamentals previously shown to predict aggregate activity: (i) special items (Hann et al., 2021; Abdalla and Carabias, 2022) and (ii) forecasted earnings growth (Choi et al., 2016). Specifically, we test whether current annual special items and forecasted earnings growth (based on analyst estimates) from firms in the top *MRisk* quartile better forecast future GDP growth than those from the bottom quartile.[15] In both tests, we benchmark *MRisk* against each of the previously analyzed alternative proxies by sorting firms into quartiles based on those measures and using quartile-level averages of special items and forecasted earnings growth to predict GDP in year t+1. Finally, we evaluate predictive power using R-squared from OLS regressions. As reported in Table 5, *MRisk*-based sorting produces strong and monotonic increases in explanatory power across both predictors. For example, when forecasting nominal GDP growth, the R-squared rises from 0.25 to 0.59 for special items, and from 0.07 to 0.53 for analyst forecasts—suggesting that *MRisk* is an effective sorting variable for identifying firms that help forecast future macroeconomic conditions.

[15] Abdalla and Carabias (2022) find that after accounting for serial correlation in GDP growth, only special items (as opposed to corporate earnings) predictive future changes in the macroeconomy. Therefore, we do not examine the ability of the identified firms' earnings to predict future economic activity. Furthermore, we do not examine management forecasts since the propensity to provide earning's guidance has declined substantially over time (e.g., Ali et al., 2023).

VI. WHAT ECONOMIC FORCES DRIVE SENSITIVITY TO AGGREGATE SHOCKS?

Decomposing *MRisk* Into Individual Macro Topics

We decompose *MRisk* into key macroeconomic topics. This step is essential to assess whether macro disclosures provide unique insights into the economic drivers underlying firms' sensitivity to aggregate shocks—information we believe is not captured by existing measurement approaches. Following prior literature, we identify core macroeconomic risks as defined by Chen et al. (1986, p. 387) and Bonsall et al. (2013, p. 231), who classify macro factors into six conceptual categories: (i) aggregate output, (ii) credit markets, (iii) prices, (iv) housing, (v) interest rates, and (vi) global trade. Accordingly, we isolate from *MRisk* the unigrams and bigrams most closely associated with each of these categories.

Figure 3 plots the average level and cross-sectional standard deviation (i.e., dispersion) of firm-level macro topics over time. If *MRisk* captures unique information about the economic drivers of systematic risk exposure, we would expect these disclosed macro topics to exhibit meaningful variation over time in line with underlying economic trends.

Figure 3 shows that both the level of macro topic disclosures and their cross-sectional dispersion vary meaningfully over time. For instance, discussions of credit markets and the dispersion in such disclosures increase sharply in late 2007 and 2008 during the financial crisis, when credit supply was constrained. As conditions stabilized, these disclosures declined. Similarly, references to aggregate output—and the dispersion across firms—rise during the Great Recession and again during the COVID-19 pandemic. In contrast, discussions of prices and inflation decline during the COVID-19 period, reflecting muted inflationary concerns at that time. Finally, the figure highlights an increase in global trade risk disclosures following a period of heightened trade tensions. Overall, the results suggest that topic-level macro disclosures track underlying macroeconomic trends and that dispersion in disclosures rises during periods of heightened aggregate risk, consistent with increased cross-firm heterogeneity in macroeconomic risk exposure.

Macro Topics' Usefulness to Understanding Ex-ante Exposure to Aggregate Shocks

Our analyses show that macro disclosures can be effectively used to measure firms' future performance sensitivity to aggregate shocks. However, we argue that their key contribution lies in providing insight into the underlying economic drivers of such exposure. In particular, these disclosures help explain *why* certain firms are more or less sensitive to macroeconomic shocks—offering a deeper understanding of the mechanisms behind systematic earnings risk.

Building on the analysis in Figure 3, we construct disclosure-based variables that capture the relative frequency of specific macro topics discussed in 10-K filings. We begin with a descriptive analysis of variation in macro topic salience over time. For each year, we rank the six topics based on their explanatory power for *MRisk*, with results reported in Table 6. The findings reveal that macroeconomic shocks are heterogeneous and that the relevance of underlying drivers varies across years. Notably, macro disclosures appear to anticipate key shocks before they manifest in observable macroeconomic outcomes. For example, in 2007—preceding the full onset of the financial crisis—the most salient topics are credit markets and real estate, consistent with the early stages of the crisis. In 2008, as the crisis broadened, credit markets and aggregate output emerge as the most relevant. Similarly, the increasing prominence of global trade risk from 2017 to 2020 coincides with major political developments and COVID-related supply chain disruptions. While the interpretation draws on hindsight for validation, it is important to note that our disclosure-based measure is constructed *ex ante* using only publicly available 10-K and WSJ information.

We proceed to confirm our conjecture that the reason *why* our measure captures ex ante time-varying exposure to aggregate shocks is that it reflects the changing nature of both the shocks themselves and firms' exposures to them—before such shocks materialize in observable macroeconomic outcomes. Therefore, we construct two alternative versions of *MRisk*: one that includes only macroeconomic terms related to the most relevant macro topic for a given year (*MRisk_Top_Topic*), based on the rankings presented in Table 6, and another based on the least relevant topic (*MRisk_Bottom_Topic*). While multiple topics may contribute to earnings sensitivity to aggregate shocks, we expect this sensitivity to be primarily driven by the top-ranked topic.

Table 7 supports this expectation. Specifically, we find that firms' sensitivity to aggregate shocks is largely explained by the most salient macro topic disclosed in the 10-K for a given year, and this leading topic varies over time. Together with the results in Table 2 and Figure 3—showing substantial variation in macro disclosures across and within firm—these findings underscore the core mechanism behind our measure's effectiveness: both the nature of macroeconomic shocks and firm-specific exposures to these shocks evolve over time and both are captured by macro disclosures.

Overall, Tables 6 and 7 confirm that macroeconomic disclosures not only capture time variation in aggregate shocks *before* these shocks manifest in observable macroeconomic outcomes, but also provide insights into the *type* of macroeconomic risk and the firm's exposure to it—reinforcing the economic interpretability and predictive value of our approach.

VII. DO MACRO DISCLOSURES CAPTURE PRICED SYSTEMATIC RISK?

As noted above, the documented relationship between macroeconomic disclosures and macro sensitivity suggests that such disclosures may also inform investors about firms' systematic risk exposure—*provided these macroeconomic exposures are priced*. If macro disclosures convey information about priced systematic risk, we expect two empirical patterns. First, there should be a positive association between *MRisk* and future stock returns, indicating that the information is priced—that is, firms more sensitive to macroeconomic shocks earn higher expected returns as compensation for higher non-diversifiable risk (e.g., Fama-MacBeth, 1973). Second, firms with higher macro sensitivity should exhibit stronger loadings (betas) on a factor-mimicking portfolio constructed using *MRisk*-based sorts, consistent with the theoretical and empirical framework in Fama and French (1996).

We test these predictions in Tables 8 and 9. Table 8 (Panel A) sorts firms into four portfolios based on *MRisk* and reports average monthly excess returns for the twelve months following the 10-K filing. Returns increase monotonically across the *MRisk* distribution, from 0.38% in the lowest quartile to 0.82% in the highest—suggesting that *MRisk* is associated with priced risk. Table 9 provides further evidence using firm-level regressions. Both pooled OLS specifications (with firm and time fixed effects

and standard errors clustered by month) and Fama-MacBeth (1973) cross-sectional regressions confirm a positive and statistically significant relationship between *MRisk* and future stock returns.[16]

To assess whether this priced risk corresponds to *systematic* risk, we construct factor-mimicking portfolios using high-minus-low *MRisk*-based returns. If *MRisk* captures systematic risk, firms with greater macro sensitivity should exhibit higher betas with respect to this portfolio. Table 8 (Panel B) reports factor loadings from time-series regressions of *MRisk*-sorted portfolio returns on the mimicking factor. The loadings increase monotonically from –0.79 to 0.21, consistent with *MRisk* capturing priced systematic risk. Additionally, Table 8 (Panel C) presents second-stage Fama-MacBeth regressions, where cross-sectional portfolio returns are regressed on these factor loadings. The positive and significant coefficient on the beta term indicates a risk premium for macro sensitivity, reinforcing the interpretation of *MRisk* as a measure of systematic risk.

We also benchmark *MRisk* against the existing proxies examined before. Notably, *MRisk* is the <u>only</u> measure that generates a monotonic pattern in both portfolio returns (Table 8, Panel A) and factor loadings (Table 8, Panel B). The earnings beta measure proposed by Ellahie (2021) performs second best, exhibiting some pricing relevance. However, *MRisk* consistently reflects priced systematic risk, *in addition to* offering greater breadth and interpretability in its information content—dimensions that are not captured by the earnings beta.

Overall, these analyses suggest that macroeconomic disclosures contain forward-looking information about firms' sensitivity to macroeconomic shocks, that this sensitivity varies over time, that its economic drivers can be understood, ex ante, through the disclosures, and that such time-varying exposure to aggregate risk is priced in stock returns.

VIII. CONCLUSION

In this paper, we develop a simple, interpretable, and objective textual measure that captures a company's propensity to provide macroeconomic discussions within 10-K reports. We demonstrate that macroeconomic disclosures enable users to (i) identify companies whose future performance is highly

[16] Our results are similar when ten portfolios, rather than four, are used to examine future stock returns.

correlated with future economic conditions (i.e., firms with higher levels of macro sensitivity), (ii) understand the economic drivers underlying such aggregate risk exposure; and (iii) capture priced systematic risk. Overall, our findings offer new evidence about the fundamental role of accounting disclosures in identifying and explaining firm-level time-varying systematic risk.

On one hand, our findings contribute to the literature that uses accounting information to measure and understand systematic risk in earnings (e.g., Ball and Brown, 1967, 1968; Beaver et al., 1970; Penman and Zhang, 2020; Ellahie, 2021). We show that macroeconomic disclosures provide a powerful tool for better quantifying firms' earnings betas and interpreting systematic risk *ex ante*, before aggregate shocks materialize—an empirical challenge that prior studies have struggled to address.

On the other hand, our analysis speaks to concerns in the empirical asset pricing literature regarding the proliferation of hard-to-interpret risk factors (Cochrane, 2011; Hou et al., 2020; Bessembinder et al., 2025)—the so-called "factor zoo." We demonstrate that firm disclosures can offer interpretable, *ex ante* priced risk factors rooted in economic fundamentals.

Finally, this study complements the accounting and finance literature on risk factor disclosures (Kravet and Muslu, 2013; Campbell et al., 2014; Hassan et al., 2019, 2023; Konchitchki and Xie, 2023), which has not traditionally focused on showing how firm-level disclosures help explain macro-related earnings sensitivity and priced systematic risk.

We envision several applications and avenues for future research using the disclosure-based measure we develop. Naturally, future work can use macro disclosure to better identify bellwether firms within capital market and accounting research settings. Furthermore, macro disclosure can help examine strategic disclosure choices within regulatory filings—for instance, conditioning the analysis of macroeconomic discussions on the sign of the news. Finally, we believe our measure can shed further light on the asset pricing implications of firms' exposure to time-varying aggregate shocks.

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APPENDIX A – Descriptions of Primary Variables

Aggregate Economic Variables		
Δ *nGDP*	Yearly change in nominal U.S. Gross Domestic Product downloaded from Table 1.1.5 of the Bureau of Economic Analysis (BEA)	BEA
Δ *rGDP*	Yearly change in real U.S. Gross Domestic Product downloaded from Table 1.1.6 of the Bureau of Economic Analysis (BEA)	BEA
Δ *Inflation*	Yearly change in U.S. inflation downloaded from Table 1.1.9 of the Bureau of Economic Analysis (BEA)	BEA
Experimental Variable		
MRisk	Ratio of macroeconomic terms (as defined within Appendix B) to total words within a 10-K filing in year *t*; multiplied by 1,000	EDGAR
Firm Fundamentals		
Δ *OI*	Firm-level changes in operating income for year *t*, scaled by beginning of the year stock price.	Compustat and CRSP
Alternative Proxies		
Numbers-Based		
Size	Natural logarithm of a firm's market capitalization	CRSP
Market Beta	Coefficient on the market factor computed using a Fama-French three-factor model estimated with daily returns over the past 12 months	CRSP
Earnings Beta	A firm-level coefficient calculated using a trailing, or backward-rolling-window, approach. This involves regressing the yearly changes in operating income, which are scaled by the previous year's stock prices, against changes in nominal GDP. Only firms that have a minimum of 5 observations are included.	Compustat BEA
Bell	Indicator equal to 1, 2, 3, or 4 based on the quartile pooled rank of a firm's R^2 from the Bonsall et al. (2013, p. 231) regression, using their vector of macroeconomic variables.	Compustat
Text-Based		
RFD_{CEA}	Ratio of Campbell et al. (2014) risk words, in Items 1A and 7, to total words within a 10-K filing in year *t*; multiplied by 1,000.	EDGAR
$RFD_{CEA\ SYST}$	Ratio of Campbell et al. (2014) systematic risk words, in Items 1A and 7, to total words within a 10-K filing in year *t*; multiplied by 1,000.	EDGAR
$RFD_{CEA\ IDIO}$	Ratio of Campbell et al. (2014) idiosyncratic risk words, in Items 1A and 7, to total words within a 10-K filing in year *t*; multiplied by 1,000.	EDGAR
RFD_{KM}	Ratio of Kravet and Muslu (2013) risk words, in Items 1A and 7, to total words within a 10-K filing in year *t*; multiplied by 1,000.	EDGAR
RFD_{LM}	Ratio of Loughran and McDonald (2011) uncertainty words to total words within a 10-K filing in year *t*; multiplied by 1,000	EDGAR
PRD	Ratio of top 120 political risk bigrams identified by Hassan et al. (2019) to total words within a 10-K filing in year *t*; multiplied by 1,000	EDGAR

APPENDIX B - Macroeconomic Terms included in *MRisk*

	Percentile Rank	
Unigrams	*10-K Sample*	*WSJ Sample*
Macro, Macroeconomic/s, Macroeconomy	Top 1%	Top 1%
Import/ing/ed, Export/ing/ed	Top 1%	Top 1%
Inflation, Deflation	Top 1%	Top 1%
GDP, GNP	Top 1%	Top 1%
Recession	Top 1%	Top 1%
Currency	Top 1%	Top 1%
FED	Top 1%	Top 1%
Bigrams	*10-K Sample*	*WSJ Sample*
Economic condition/environment/downturn/factor/trend/ growth/activity/development/slowdown/instability/ uncertainty/recovery/climate/data/cycle/ crisis/indicator/output/expansion	Top 1%	Top 1%
Capital/credit/global/international/exchange/emerging/bear/ bull market	Top 1%	Top 1%
Market/credit/global/international/exchange/economic/ risk	Top 1%	Top 1%
Global/international/emerging/general economy	Top 1%	Top 1%
Foreign exchange/investor/investment	Top 1%	Top 1%
Real estate/property/growth/rate	Top 1%	Top 1%
Federal reserve, Central bank	Top 1%	Top 1%
Gross domestic/national	Top 1%	Top 1%
Monetary/fiscal policy	Top 1%	Top 1%
Interest/discount rate	Top 1%	Top 1%
Business cycle	Top 1%	Top 1%
Global trade	Top 1%	Top 1%

This table reports the macroeconomic terms included in the text-based disclosure measure. These terms represent the intersection of the top 1% most frequent macroeconomic unigrams and bigrams (i.e., individual words and pairs of consecutive words, respectively, excluding common "stop words") in (i) a sample of 86,624 10-K reports filed between 1994 and 2021, and (ii) a sample of 22,993 *Wall Street Journal* "Economic News" articles published between 1994 and 2021. The 10-K reports are downloaded from SEC EDGAR while the news articles from *Dow Jones Factiva* and exclude documents with fewer than 5 sentences and 100 words. Exact matching is performed for all the listed terms except for "Inflation", "Deflation", "Recession", and "Currency" that are matched as substrings (to capture words such as "Hyperinflation" or "Recessionary"). While not listed, plural tenses are searched for when relevant. 10-K reports, *WSJ* articles, and macro-related terms are all converted into lower case before matching.

APPENDIX C - Examples of Macroeconomic Disclosures

Caterpillar Inc – CIK 0000018230
Report year: 2018, Item 7 – Management Discussion and Analysis

Retail revenue for 2018 was $1.31 billion, an increase of $73 million from 2017. The increase was due to a $73 million favorable impact from higher **interest rates** on retail finance receivables. For the year ended December 31, 2018, retail average earning assets were $23.10 billion, an increase of $9 million from 2017.

The average yield was 5.66 percent for 2018, compared with 5.35 percent in 2017. Operating lease revenue for 2018 was $1.01 billion, an increase of $26 million from 2017. The increase was due to a $27 million favorable impact from higher average rental rates on operating leases, partially offset by a $1 million unfavorable impact from lower average earning assets.

Wholesale revenue for 2018 was $415 million, an increase of $108 million from 2017. The increase was due to an $83 million favorable impact from higher average earning assets and a $25 million favorable impact from higher **interest rates** on wholesale finance receivables. For the year ended December 31, 2018, wholesale average earning assets were $4.85 billion, an increase of $1.04 billion from 2017. The average yield was 8.55 percent for 2018, compared with 8.04 percent in 2017.

Apple Inc. – CIK 0000320193
Report year: 2019, Item 1A – Risk Factors

Global and regional economic conditions could materially adversely affect the Company's business, results of operations, financial condition and growth.

The Company has international operations with sales outside the U.S. representing a majority of the Company's total net sales. In addition, a majority of the Company's supply chain, and its manufacturing and assembly activities, are located outside the U.S. As a result, the Company's operations and performance depend significantly on global and regional **economic conditions**.

Adverse **macroeconomic** conditions, including **inflation**, slower growth or **recession**, new or increased tariffs, changes to fiscal and **monetary policy**, tighter credit, higher **interest rates**, high unemployment and **currency** fluctuations could materially adversely affect demand for the Company's products and services. In addition, consumer confidence and spending could be adversely affected in response to financial market volatility, negative financial news, conditions in the real estate and mortgage markets, declines in income or asset values, changes to fuel and other energy costs, labor and healthcare costs and other **economic factors**.

In addition to an adverse impact on demand for the Company's products, uncertainty about, or a decline in, global or regional **economic conditions** could have a significant impact on the Company's suppliers, contract manufacturers, logistics providers, distributors, cellular network carriers and other channel partners. Potential effects include financial instability; inability to obtain credit to finance operations and purchases of the Company's products; and insolvency.

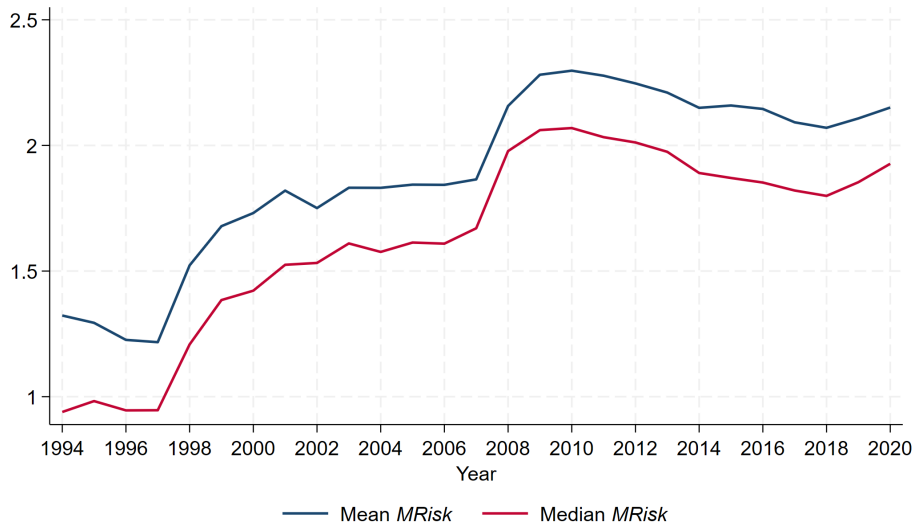
A downturn in the **economic environment** could also lead to increased credit and collectibility risk on the Company's trade receivables; the failure of derivative counterparties and other financial institutions; limitations on the Company's ability to issue new debt; reduced liquidity; and declines in the fair value of the Company's financial instruments. These and other **economic factors** could materially adversely affect the Company's business, results of operations, financial condition and growth.

The Walt Disney Company – CIK 0001744489
Report year: 2020, Item 7 – Management Discussion and Analysis

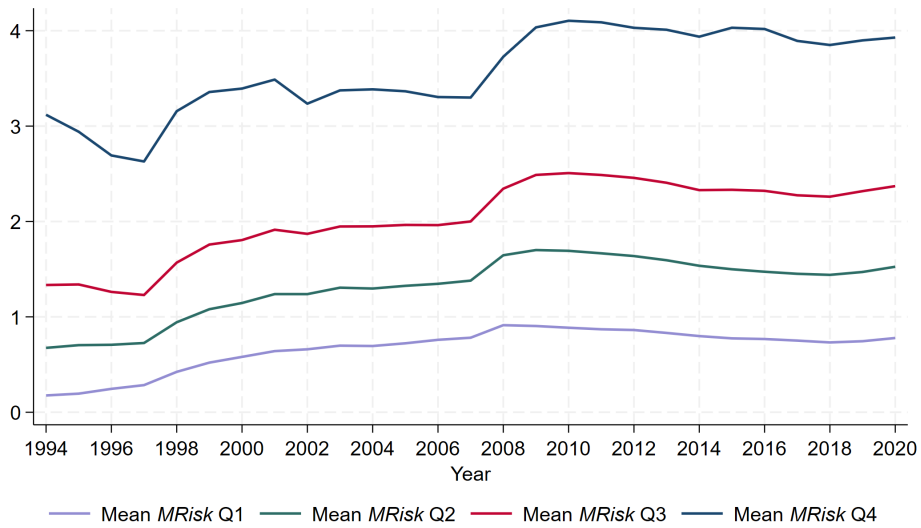
Operating expenses include operating labor, which decreased $1,304 million from $6,174 million to $4,870 million, cost of sales and distribution costs, which decreased $716 million from $2,918 million to $2,202 million, and infrastructure costs, which decreased $47 million from $2,469 million to $2,422 million. The decrease in operating labor was due to furloughs in response to lower volumes and the benefit of government credits for certain employee costs, partially offset by **inflation** and new guest offerings. Lower cost of goods sold and distribution costs were due to lower volumes. The decrease in infrastructure costs was due to lower operations support costs reflecting reduced volumes, partially offset by the write-down of assets at our retail stores and higher costs for new guest offerings. Other operating expenses, which include costs for such items as supplies, commissions/fees and entertainment offerings, decreased $463 million, from $2,454 million to $1,991 million, due to lower volumes, partially offset by higher charges for capital project abandonments.

FIGURE 1

Panel A: Time-trend of macroeconomic disclosures within 10-K reports



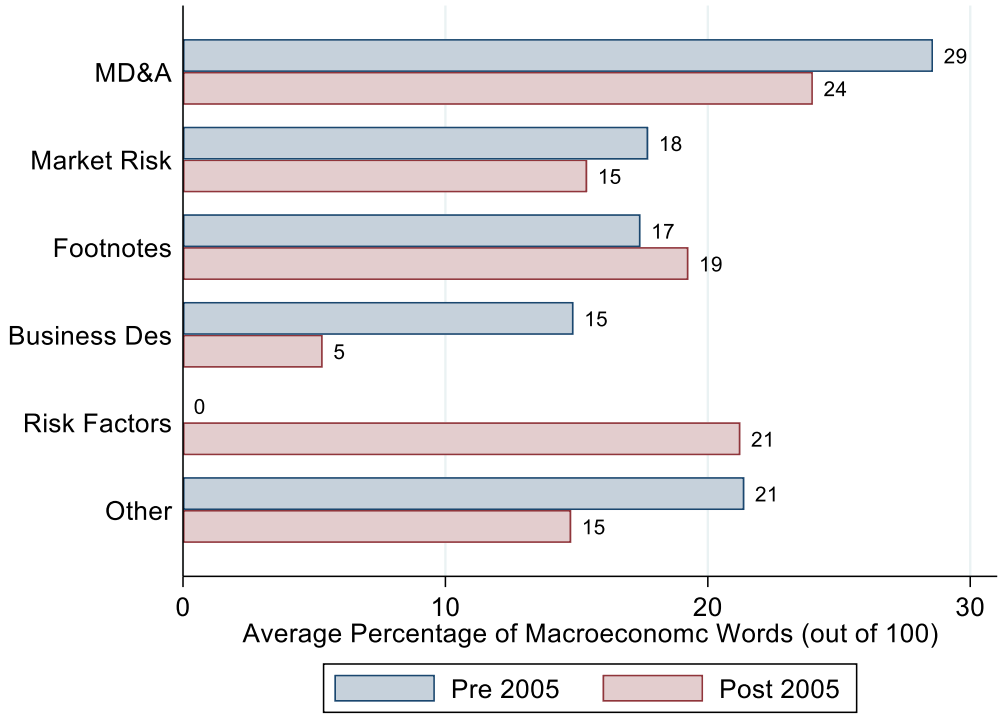
Panel B: Time-trend of macroeconomic disclosures within 10-K reports by quartile



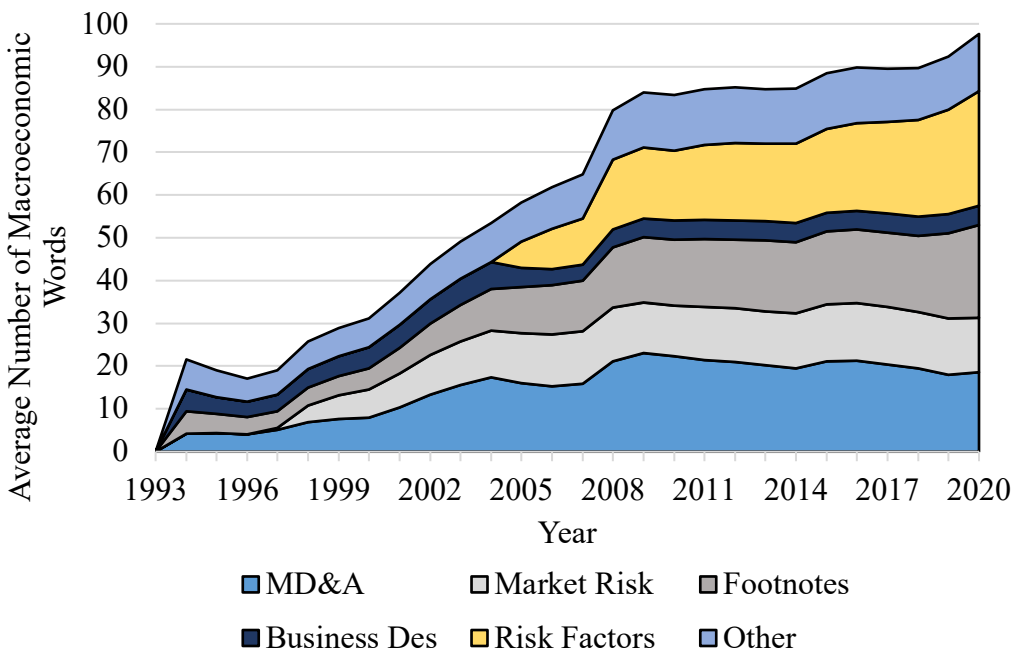
This figure reports the time-trend of macroeconomic disclosures within 10-K reports. The sample includes 86,624 firm-year observations, ranging from reporting years 1994-2020. *MRisk* is defined as the total number of macroeconomic words per thousand words in firm *i*'s 10-K filing in year *t*. Panel A reports aggregate means and medians for each year. Panel B shows means for each year by quartile of *MRisk*.

FIGURE 2

Panel A: Location of macroeconomic disclosures within 10-K reports



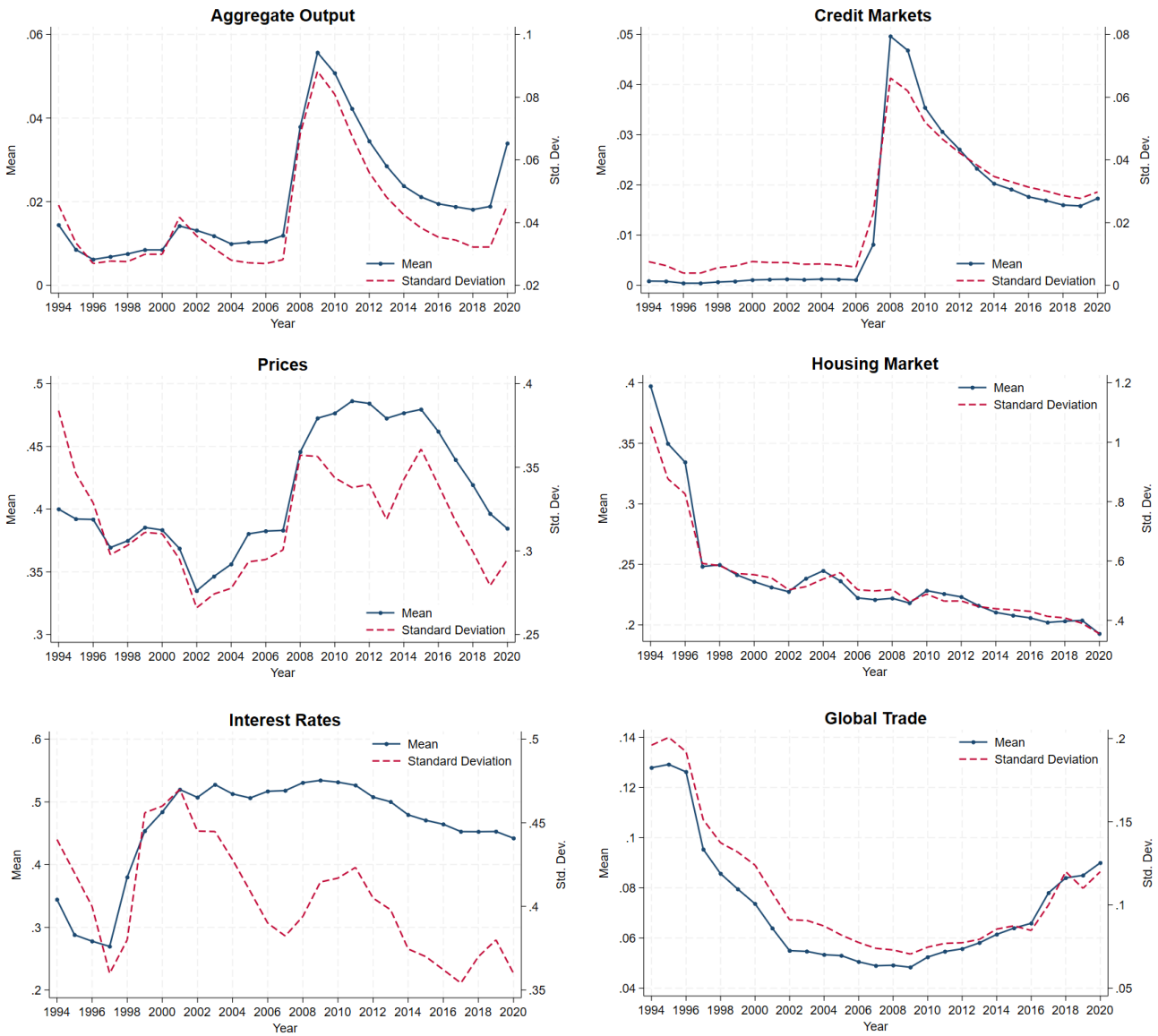
Panel B: Location of macroeconomic disclosures within 10-K reports over time



This figure reports the location of macro disclosures within specific sections of the 10-K. The sample includes 86,624 firm-year observations, during reporting years 1994-2020. Panel A shows the average percentages of macroeconomic words in each section, for the top five sections based on the frequency of macroeconomic terms. Light-blue bars report averages prior to the introduction of the "Risk Factor" section by the SEC (i.e., before 2005), while light-red bars show averages after 2005. Panel B displays the time trend of the average number of macroeconomic words in each section. "Other" includes all other 10-K sections.

FIGURE 3

Time-series frequency and dispersion of select macroeconomic topics



This figure presents the time-series of scaled counts of macroeconomic terms per 1,000 words in firm *i*'s 10-K for year *t*, categorized into three groups: (1) "Aggregate Output," (2) "Credit Markets," (3) "Prices," (4) "Housing Market," (5) Interest Rates," and (6) Global Trade." For each topic, the figure plots the annual average count of related terms (solid line) and the corresponding cross-sectional standard deviation (dashed line), reflecting the dispersion across firms. The topics are defined as follows: "Aggregate Output" includes terms such as *economic conditions*, *recession*, *economic trends*, and *growth rates*; "Credit Markets" *includes credit market(s) and credit risk(s);* "Prices" *includes price pressure, prices, and inflation;* "Housing Market" includes *real estate*; "Interest Rates" *includes interest rate(s)*, *discount rate(s)*, *federal reserve*, *monetary policy*, and *fiscal policy*; and "Global Trade" includes *imports* and *exports*.

TABLE 1
Sample Selection and Descriptives

Panel A: Selection Criteria

	Number
EDGAR 10-K filings by companies reporting between 1994 and 2020, with valid CIK, GVKEY, and PERMNO, and minimum 500 words	143,367
Less: 10-K reports:	
- filed by financial institutions (SIC codes 6000-6999)	(32,071)
Total EDGAR 10-K filings with available macroeconomic disclosures	111,296
Less: observations with missing:	
- CRSP, Compustat, I/B/E/S identifiers	(17,195)
- lagged disclosures to compute $MRisk_{t-1}$	(7,477)
Total firm-years included in the primary sample	**86,624**

Panel B: Summary Statistics

	N	Mean	SD	P25	P50	P75
MRisk	86,624	1.90	1.22	0.98	1.64	2.54
Size	86,624	5.76	2.31	4.09	5.77	7.36
Market Beta	86,624	0.92	0.53	0.60	0.91	1.21
Earnings Beta	72,360	2.20	32.78	-4.68	2.27	9.60
Bell	86,624	2.17	1.32	1.00	2.00	3.00
RFD_{CEA}	86,624	16.38	10.00	8.49	16.67	24.03
$RFD_{CEA\ SYST}$	86,624	4.38	3.55	1.75	3.91	6.04
$RFD_{CEA\ IDIO}$	86,624	12.69	7.35	7.18	12.83	18.08
RFD_{KM}	86,624	10.29	4.06	7.11	10.30	13.16
RFD_{LM}	86,624	13.26	3.65	10.60	13.39	15.92
PRD	86,624	11.25	2.92	9.21	10.70	12.66

This table presents the sample formation process and descriptive statistics for the primary sample of 86,624 firm-year observations, ranging from reporting years 1994-2020. Panel A reports the sample formation criteria. Panel B presents summary statistics. All variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles of their yearly distribution.

TABLE 2
Time, Industry, and Firm-level Variation of *MRisk*

Panel A: Variance Decomposition

A.1 *Incremental R^2*	*2-digit SIC* (1)	*3-digit SIC* (2)	*4-digit SIC* (3)
Time FE	5.9%	5.9%	5.9%
Industry FE	3.7%	12.3%	15.0%
Year x Industry FE	9.3%	12.0%	13.5%
Implied Firm-level	81.1%	69.8%	65.6%
Sum	*100.0%*	*100.0%*	*100.0%*
Number of Sectors	64	248	398

A.2 *Analysis of Firm-level Variation*	*Permanent (Firm FE)* (1)	*Time-Varying (Residual)* (2)
R^2	59.6%	40.4%

Panel B: Transition Matrix

	MRisk $_t$ Quartile				
MRisk $_{t-1}$ Quartile	*Q1*	*Q2*	*Q3*	*Q4*	*Total*
Q1	75.1%	17.8%	4.6%	2.5%	100.0%
Q2	18.2%	59.7%	16.5%	5.6%	100.0%
Q3	5.4%	19.2%	63.4%	12.0%	100.0%
Q4	2.3%	4.3%	15.0%	78.4%	100.0%

This table presents the variability of *MRisk* over time, across industries, and within firms. Panel A reports the results of a variance decomposition analysis in which *MRisk* is regressed on various fixed effects structures, with incremental R^2 values reported. Columns (1)-(3) in Panel A.1 are based on three alternative definitions of industries/sectors: 2-digit, 3-digit, and 4-digit SIC codes. Panel A.2 highlights the proportion of firm-level variation that changes over time within firms, calculated as the residual R^2 from regressing *MRisk* on firm fixed effects. Panel B presents the transition matrix of firms across quartiles of *MRisk* between year *t*-1 and year *t*. The sample comprises 86,624 firm-year observations, spanning the reporting years 1994 to 2020.

TABLE 3
Portfolio-level Sensitivity to Future Macroeconomic Changes

Sorting Variables $_{i,t-1}$	Change in Macroeconomic Activity $_t$																								Monotonic Trend
	Nominal Output: Δ *nGDP*								Real Output: Δ *rGDP*								Prices: Δ *Inflation*								
	Coefficient				R^2				*Coefficient*				R^2				*Coefficient*				R^2				
	Q1	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	
Experimental Variable $_{i,t-1}$																									
MRisk	**0.02**	**0.04**	**0.07**	**0.12**	**0.09**	**0.14**	**0.20**	**0.40**	**0.03**	**0.04**	**0.08**	**0.13**	**0.09**	**0.12**	**0.18**	**0.32**	**0.05**	**0.09**	**0.15**	**0.34**	**0.02**	**0.07**	**0.08**	**0.25**	**Yes**
p-value Q4-Q1	*<0.00****				*<0.00****				*<0.00****				*<0.00****				*<0.00****				*<0.00****				
Alternative Macro Proxies $_{i,t-1}$																									
Numbers-based																									
Size	0.00	0.00	0.00	0.25	0.06	0.01	0.08	0.32	0.00	0.00	0.00	0.27	0.04	0.03	0.08	0.27	0.00	0.00	0.01	0.62	0.08	0.02	0.02	0.16	No
Market Beta	0.01	0.06	0.10	0.08	0.00	0.27	0.33	0.23	0.01	0.07	0.11	0.09	0.02	0.24	0.28	0.18	-0.01	0.13	0.25	0.25	0.00	0.10	0.17	0.17	No
Earnings Beta	0.03	0.13	0.16	0.05	0.04	0.47	0.37	0.28	0.04	0.15	0.17	0.06	0.06	0.48	0.29	0.25	0.06	0.27	0.45	0.14	0.01	0.17	0.24	0.16	No
Bell	0.09	0.04	0.06	0.10	0.39	0.14	0.34	0.23	0.09	0.04	0.06	0.11	0.34	0.12	0.31	0.19	0.21	0.08	0.11	0.25	0.19	0.06	0.11	0.11	No
Text-based																									
RFD_{CEA}	0.06	0.03	0.03	0.11	0.33	0.12	0.10	0.28	0.07	0.03	0.03	0.12	0.32	0.09	0.08	0.22	0.12	0.08	0.09	0.30	0.09	0.08	0.06	0.18	No
$RFD_{CEA\ SYST}$	0.04	0.01	0.04	0.12	0.25	0.00	0.14	0.37	0.05	0.01	0.05	0.13	0.23	0.01	0.11	0.31	0.09	0.04	0.11	0.33	0.09	0.02	0.08	0.21	No
$RFD_{CEA\ IDIO}$	0.10	0.06	0.03	0.04	0.49	0.21	0.10	0.20	0.11	0.06	0.04	0.04	0.43	0.16	0.09	0.15	0.21	0.18	0.06	0.13	0.20	0.15	0.03	0.15	No
RFD_{KM}	0.10	0.05	0.05	0.06	0.46	0.29	0.18	0.11	0.10	0.05	0.06	0.06	0.40	0.25	0.26	0.09	0.23	0.11	0.14	0.14	0.21	0.14	0.10	0.05	No
RFD_{LM}	0.11	0.07	0.03	0.04	0.50	0.23	0.11	0.12	0.13	0.08	0.03	0.04	0.45	0.19	0.09	0.09	0.25	0.18	0.08	0.12	0.19	0.12	0.05	0.10	No
PRD	0.04	0.04	0.08	0.09	0.17	0.23	0.40	0.13	0.04	0.04	0.09	0.10	0.14	0.17	0.37	0.11	0.11	0.13	0.17	0.21	0.10	0.18	0.14	0.06	No
N	*27 yearly observations*								*27 yearly observations*								*27 yearly observations*								

This table presents a portfolio-level univariate validation analysis of *MRisk*. Specifically, the following regression is estimated across different levels of *MRisk* and ten alternative macroeconomic information proxies—four numbers-based measures (*Size*, *Market Beta*, *Earnings Beta*, and *Bell*) and six text-based indicators (RFD_{CEA}, $RFD_{CEA\ SYST}$, $RFD_{CEA\ IDIO}$, RFD_{KM}, RFD_{LM}, and *PRD*).

$$\Delta OI_{p,t} = \beta_0 + \beta_1\, \Delta\, Macroeconomic\ Activity_t + \varepsilon_t$$

where $\Delta OI_{p,t}$ is the value-weighted cross-sectional average of all firm-level changes in operating income for year *t*, scaled by beginning of the year stock price. Observations are sorted into quartile portfolios each year based on the *lagged* level of the previous macroeconomic information proxies; the total number of observations in each time-series regression is equal to 27. Δ *Macroeconomic Activity* is alternatively (i) $\Delta\, nGDP_t$, the annual nominal GDP growth rate for year *t,* (ii) $\Delta\, rGDP_t$, the annual real GDP growth rate for year *t*, (iii) $\Delta\, Inflation_t$, the annual inflation growth rate for year *t*. All variables are described in Appendix A. Differences in coefficients and R^2 are tested using a bootstrapping protocol with replacement and 500 iterations. ***, **, * indicate significance at <0.01, <0.05, <0.10 (two-tailed tests).

TABLE 4
Firm Performance Sensitivity to Future Macroeconomic Changes

	Dependent Variable: $\Delta OI_{i,t}$					
	Full Sample			>5 Observations per Firm		
Δ *Macro* $_t$	Δ *nGDP*	Δ *rGDP*	Δ *Inflation*	Δ *nGDP*	Δ *rGDP*	Δ *Inflation*
Independent Variables $_{i,t-1}$	(1)	(2)	(3)	(4)	(5)	(6)
MRisk* x Δ *Macro	**2.31*** **	**2.10*** **	**1.12** **	**2.08*** **	**1.83*** **	**1.18** **
	(0.00)	**(0.00)**	**(0.04)**	**(0.00)**	**(0.00)**	**(0.03)**
Size x Δ *Macro*	1.64	0.95	2.32**	1.13	0.59	1.89*
	(0.11)	(0.32)	(0.02)	(0.29)	(0.57)	(0.06)
Market Beta x Δ *Macro*	-0.02	-0.05	-0.30	-0.03	0.27	-0.63
	(0.70)	(0.92)	(0.49)	(0.95)	(0.67)	(0.20)
Earnings Beta x Δ *Macro*				2.21	2.36*	2.43*
				(0.11)	(0.10)	(0.08)
$RFD_{CEA\ SYST}$ x Δ *Macro*	1.10	0.59	0.96	0.46	-0.04	1.56
	(0.29)	(0.59)	(0.13)	(0.68)	(0.97)	(0.10)
$RFD_{CEA\ IDIO}$ x Δ *Macro*	-0.36	0.09	-0.43	-0.08	-0.42	-1.60
	(0.75)	(0.94)	(0.46)	(0.50)	(0.74)	(0.11)
RFD_{KM} x Δ *Macro*	-1.57	-1.31	-0.73	-0.19	0.02	-0.59
	(0.23)	(0.26)	(0.39)	(0.89)	(0.98)	(0.54)
RFD_{LM} x Δ *Macro*	0.08	-0.41	1.30	0.01	-0.48	1.36
	(0.94)	(0.69)	(0.10)	(0.99)	(0.67)	(0.12)
PRD x Δ *Macro*	0.86	0.66	0.47	0.53	0.30	0.60
	(0.13)	(0.22)	(0.31)	(0.40)	(0.63)	(0.25)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year x Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	72,934	72,934	72,934	60,124	60,124	60,124
Adjusted R^2	0.13	0.13	0.13	0.12	0.12	0.12

This table presents results from estimating the following equation:

$$\Delta OI_{i,t} = \beta_0 + \beta_1 \Delta Macro_t + \beta_2 MRisk_{i,t-1} + \beta_3 \Delta Macro_t \times MRisk_{i,t-1} + \sum_{j=4}^{J} \beta_j Alternative\ Proxies_{i,t-1}$$
$$+ \sum_{k=J+1}^{K} \beta_k \Delta Macro_t \times Alternative\ Proxies_{i,t-1} + \sum_{v=K+1}^{V} \beta_v Controls_{i,t-1}$$
$$+ \sum_{q=V+1}^{Q} \beta_q \Delta Macro_t \times Controls_{i,t-1} + \sum_{n=Q+1}^{N} \beta_n Controls_{i,t} + Firm\ FE_i + Year \times Sector\ FE_{s,t} + \varepsilon_{i,t}$$

where $\Delta OI_{i,t}$ is the firm-level change in operating income for year *t*, scaled by beginning of the year stock price. $\Delta\ Macro_t$ is alternatively the yearly change in nominal GDP, real GDP and inflation. *MRisk* is the total number of macroeconomic words per 1,000 total words in firm *i's* 10-K filing for year *t-1*. Controls include book-to-market, ROA, leverage, special items, number of geographic segments, and idiosyncratic stock returns volatility over the prior 12 months. *Bell* is excluded because time-invariant. All variables are defined in Appendix A. Columns (1)–(3) present results for the full sample, while Columns (4)–(6) report results for the subsample of firms with at least five observations, a necessary condition for calculating and including *Earnings Beta*. Industry-year dummies are defined based upon 4-digit SIC codes. OLS coefficients are standardized and multiplied by 10 to enhance readability. Continuous variables are winsorized at the 1st and 99th percentiles of their yearly distribution, except for disclosure variables, which are expressed in logarithms. All regressions include an intercept (not reported). Cluster-robust *p*-values are reported in parentheses, with standard errors clustered by firm. ***, **, and * indicate significance at the 0.01, 0.05, and 0.10 levels, respectively (two-tailed tests).

TABLE 5
***MRisk*, Firm Performance, and Future Macroeconomic Conditions**

	Change in Macroeconomic Activity $_{t+1}$																								
	Nominal Output: Δ *nGDP*								Real Output: Δ *rGDP*								Prices: Δ *Inflation*								Monotonic Trend
Predictor $_{p,t}$	*E[Δ Earnings]*				*Special Items*				*E[Δ Earnings]*				*Special Items*				*E[Δ Earnings]*				*Special Items*				
Sorting Vars $_{i,t}$	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	*Q1*	*Q2*	*Q3*	*Q4*	
R^2 of Experimental Variable																									
MRisk	**0.07**	**0.41**	**0.47**	**0.53**	**0.25**	**0.43**	**0.57**	**0.59**	**0.03**	**0.34**	**0.40**	**0.46**	**0.22**	**0.39**	**0.48**	**0.49**	**0.13**	**0.20**	**0.22**	**0.24**	**0.11**	**0.17**	**0.27**	**0.30**	**Yes**
p-value Q4-Q1	*< 0.00****				*< 0.00****				*< 0.00****				*< 0.00****				*< 0.00****				*< 0.00****				
R^2 of Alternative Macro Proxies																									
Numbers-based																									
Size	0.01	0.28	0.55	0.40	0.26	0.64	0.62	0.44	0.02	0.24	0.50	0.32	0.23	0.59	0.52	0.37	0.01	0.13	0.19	0.24	0.12	0.23	0.33	0.22	No
Market Beta	0.01	0.47	0.56	0.43	0.31	0.55	0.51	0.50	0.01	0.40	0.49	0.35	0.26	0.50	0.43	0.42	0.01	0.23	0.25	0.25	0.16	0.18	0.26	0.26	No
Earnings Beta	0.32	0.32	0.54	0.47	0.48	0.53	0.49	0.42	0.25	0.28	0.47	0.39	0.41	0.47	0.41	0.34	0.21	0.13	0.24	0.26	0.22	0.23	0.25	0.25	No
Bell	0.49	0.39	0.45	0.32	0.55	0.56	0.52	0.56	0.43	0.52	0.36	0.27	0.45	0.47	0.45	0.49	0.19	0.26	0.27	0.18	0.29	0.28	0.24	0.25	No
Text-based																									
RFD_{CEA}	0.25	0.48	0.48	0.49	0.51	0.56	0.51	0.47	0.23	0.45	0.42	0.37	0.46	0.51	0.44	0.37	0.08	0.14	0.22	0.33	0.19	0.21	0.24	0.30	No
$RFD_{CEA\ SYST}$	0.03	0.40	0.57	0.50	0.47	0.57	0.53	0.44	0.01	0.40	0.54	0.39	0.40	0.50	0.49	0.35	0.07	0.08	0.17	0.31	0.22	0.25	0.18	0.26	No
$RFD_{CEA\ IDIO}$	0.45	0.48	0.53	0.36	0.48	0.48	0.55	0.55	0.40	0.40	0.44	0.28	0.45	0.41	0.46	0.46	0.19	0.23	0.29	0.23	0.15	0.24	0.27	0.29	No
RFD_{KM}	0.48	0.40	0.56	0.35	0.62	0.59	0.35	0.29	0.42	0.36	0.47	0.24	0.54	0.52	0.29	0.21	0.21	0.15	0.28	0.33	0.28	0.25	0.18	0.22	No
RFD_{LM}	0.46	0.48	0.48	0.43	0.65	0.51	0.49	0.36	0.39	0.42	0.40	0.33	0.58	0.43	0.43	0.28	0.22	0.22	0.25	0.30	0.27	0.24	0.21	0.24	No
PRD	0.41	0.41	0.53	0.43	0.46	0.58	0.52	0.52	0.34	0.36	0.49	0.33	0.38	0.50	0.44	0.46	0.21	0.18	0.29	0.30	0.25	0.27	0.26	0.20	No
N	*27 yearly observations*								*27 yearly observations*								*27 yearly observations*								

This table presents results of estimating the following equation for quartiles of alternative macroeconomic information proxies:

$$\Delta\, Macroeconomic\ Activity_{t+1} = \beta_0 + \beta_1 Predictor_{p,t} + \varepsilon_{t+1}$$

Economic Activity, computed for year *t*+1, is measured as the annual growth rate in nominal GDP, real GDP, or inflation. *Predictors*, computed in year *t*, include the cross-sectional average of firm-level differences between the consensus analyst forecast of annual EPS for year *t*+1 and realized EPS for year *t* (i.e., *E[Δ Earnings]*), and the cross-sectional average of special items scaled by market capitalization (i.e., *Special Items*). Each year, observations are sorted into quartile portfolios based on the year *t* level of macroeconomic information proxies, and OLS R^2 values are reported by quartile. The total number of observations in each time-series regression is equal to 27. All other variables are defined in Appendix A. Differences in R^2 are tested using a bootstrapping procedure with replacement and 500 iterations. ***, **, and * indicate significance at the 0.01, 0.05, and 0.10 levels, respectively (two-tailed tests).

TABLE 6
Macroeconomic Topics Contributing to *MRisk* Variation Each Year

Year	Rank 1 [*Top*]	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6 [*Bottom*]
1994	Aggregate Output	Global Trade	Interest Rates	Prices	Credit Markets	Real Estate
1995	Credit Markets	Prices	Aggregate Output	Interest Rates	Global Trade	Real Estate
1996	Interest Rates	Real Estate	Credit Markets	Aggregate Output	Prices	Global Trade
1997	Prices	Interest Rates	Credit Markets	Aggregate Output	Global Trade	Real Estate
1998	Interest Rates	Aggregate Output	Credit Markets	Global Trade	Prices	Real Estate
1999	Interest Rates	Prices	Credit Markets	Aggregate Output	Real Estate	Global Trade
2000	Interest Rates	Credit Markets	Aggregate Output	Real Estate	Prices	Global Trade
2001	Aggregate Output	Interest Rates	Credit Markets	Real Estate	Global Trade	Prices
2002	Aggregate Output	Interest Rates	Real Estate	Credit Markets	Global Trade	Prices
2003	Real Estate	Prices	Interest Rates	Credit Markets	Global Trade	Aggregate Output
2004	Prices	Real Estate	Credit Markets	Global Trade	Interest Rates	Aggregate Output
2005	Prices	Interest Rates	Global Trade	Credit Markets	Aggregate Output	Real Estate
2006	Interest Rates	Global Trade	Aggregate Output	Credit Markets	Prices	Real Estate
2007	Credit Markets	Real Estate	Aggregate Output	Interest Rates	Global Trade	Prices
2008	Credit Markets	Aggregate Output	Real Estate	Prices	Interest Rates	Global Trade
2009	Aggregate Output	Credit Markets	Prices	Interest Rates	Real Estate	Global Trade
2010	Prices	Global Trade	Interest Rates	Aggregate Output	Real Estate	Credit Markets
2011	Global Trade	Prices	Interest Rates	Real Estate	Aggregate Output	Credit Markets
2012	Interest Rates	Prices	Global Trade	Real Estate	Credit Markets	Aggregate Output
2013	Interest Rates	Global Trade	Real Estate	Prices	Aggregate Output	Credit Markets
2014	Interest Rates	Global Trade	Prices	Credit Markets	Real Estate	Aggregate Output
2015	Global Trade	Interest Rates	Prices	Aggregate Output	Credit Markets	Real Estate
2016	Global Trade	Interest Rates	Aggregate Output	Credit Markets	Real Estate	Prices
2017	Global Trade	Real Estate	Interest Rates	Prices	Aggregate Output	Credit Markets
2018	Global Trade	Real Estate	Interest Rates	Aggregate Output	Prices	Credit Markets
2019	Interest Rates	Global Trade	Aggregate Output	Real Estate	Credit Markets	Prices
2020	Aggregate Output	Interest Rates	Credit Markets	Real Estate	Global Trade	Prices

This table reports key macroeconomic topics contributing to variation in *MRisk*. For each year between 1994 and 2020, word counts associated with each topic are aggregated, and *MRisk* is regressed on the change in topic frequency relative to the prior reporting period. The resulting R^2 values are used to rank topics by explanatory power. The topics are defined as follows: "Aggregate Output" includes terms such as *economic conditions*, *recession*, *economic trends*, and *growth rates*; "Credit Markets" includes *credit market(s) and credit risk(s);* "Prices" includes *price pressure, prices, and inflation;* "Housing Market" includes *real estate*; "Interest Rates" includes *interest rate(s)*, *discount rate(s)*, *federal reserve*, *monetary policy*, and *fiscal policy*; and "Global Trade" includes *imports* and *exports*.

TABLE 7
Topic-Level Time-Varying Sensitivity to Macroeconomic Changes

Dependent Variable $_{i,t}$	**Dependent Variable: $\Delta OI_{i,t}$**		
Δ Macro $_t$	**Δ *rGDP***		
Independent Variables $_{i,t-1}$	(1)	(2)	(3)
MRisk_Top_Topic x **Δ *Macro***	**6.95*** **		**7.03*** **
	(0.00)		**(0.00)**
MRisk_Bottom_Topic x **Δ *Macro***		**2.49*** **	**2.40***
		(0.00)	**(0.09)**
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year x Industry FE	Yes	Yes	Yes
Observations	72,934	72,934	72,934
Adjusted R^2	0.13	0.13	0.13

F-Test (MRisk_Top_Topic x Δ Macro > MRisk_Bottom_Topic x Δ Macro): *p-value < 0.01*

This table presents results from estimating the following equation:

$$\Delta OI_{i,t} = \beta_0 + \beta_1 \Delta Macro_t + \beta_2 MRisk_Top_(Bottom)_Topic_{i,t-1} + \beta_3 \Delta Macro_t \times MRisk_Top_(Bottom)_Topic_{i,t-1} + \sum_{v=K+1}^{V} \beta_v Controls_{i,t-1} + \sum_{q=V+1}^{Q} \beta_q \Delta Macro_t \times Controls_{i,t-1} + Firm\ FE_i + Year \times Sector\ FE_{s,t} + \varepsilon_{i,t}$$

where $\Delta OI_{i,t}$ is the firm-level change in operating income for year *t*, scaled by beginning of the year stock price. $\Delta Macro_t$ is the yearly change in real GDP. $MRisk_Top_(Bottom)_Topic$ is the total number of macroeconomic words per 1,000 total words in firm *i's* 10-K filing for year *t-1* computed considering *only* the macroeconomic topic that alternatively contributes the most (i.e., Top) or the least (i.e., Bottom) to variation in *MRisk* for a given year. Such topics are identified and reported within Table 6. Controls include size, book-to-market, ROA, leverage, special items, number of geographic segments, and idiosyncratic stock returns volatility over the prior 12 months. Additional macroeconomic proxies are excluded from this estimation. However, the tenor of the results does not change when the latter are included. Industry-year dummies are defined based upon 4-digit SIC codes. OLS coefficients are standardized and multiplied by 10 to enhance readability. Continuous variables are winsorized at the 1st and 99th percentiles of their yearly distribution, except for disclosure variables, which are expressed in logarithms. All regressions include an intercept (not reported). Cluster-robust *p*-values are reported in parentheses, with standard errors clustered by firm. ***, **, and * indicate significance at the 0.01, 0.05, and 0.10 levels, respectively (two-tailed tests).

TABLE 8
Portfolio-level Asset Pricing Tests

Panel A: Portfolio Average Excess Return

	Monthly Excess Return $_{p,\,t+1}$ (%)									
	Equally Weighted Returns					Value Weighted Returns				
Firms Sorted by:	*Q1*	*Q2*	*Q3*	*Q4*	*Q4-Q1*	*Q1*	*Q2*	*Q3*	*Q4*	*Q4-Q1*
MRisk	**0.36**	**0.69**	**0.88**	**1.03**	**0.67*** **	**0.38**	**0.64**	**0.71**	**0.82**	**0.44*** **
Size	0.69	0.69	0.81	0.75	0.06	0.54	0.60	0.69	0.61	0.07
Market Beta	0.72	0.82	0.63	0.69	-0.03	0.74	0.55	0.78	0.72	-0.02
Earnings Beta	0.64	0.84	0.89	0.81	0.17*	0.56	0.69	0.70	0.58	0.02
Bell	0.98	0.91	0.89	0.96	-0.02	0.56	0.34	0.71	0.80	0.24*
RFD_{CEA}	0.67	0.67	0.80	0.83	0.16*	0.66	0.51	0.59	0.59	-0.07
$RFD_{CEA\ SYST}$	0.69	0.65	0.85	0.78	0.09	0.70	0.45	0.57	0.57	-0.23
$RFD_{CEA\ IDIO}$	0.80	0.74	0.75	0.66	-0.24	0.68	0.52	0.58	0.50	-0.18
RFD_{KM}	0.81	0.76	0.71	0.68	-0.13	0.67	0.69	0.64	0.58	-0.09
RFD_{LM}	0.73	0.75	0.73	0.76	0.03	0.65	0.69	0.68	0.55	-0.10
PRD	0.66	0.76	0.78	0.76	0.10*	0.59	0.64	0.61	0.65	0.07

Panel B: Factor-mimicking Portfolio Loadings – Time Series Step w/ Decile Portfolios and Value-Weighted Returns

Formed on:	*D1*	*D2*	*D3*	*D4*	*D5*	*D6*	*D7*	*D8*	*D9*	*D10*
MRisk	**-0.79**	**-0.24**	**-0.20**	**-0.11**	**-0.05**	**0.03**	**0.04**	**0.11**	**0.14**	**0.21**
Size	0.98	0.70	0.45	0.32	0.16	0.10	0.02	0.02	0.05	-0.02
Market Beta	-0.47	-0.15	-0.07	-0.03	-0.07	-0.02	0.07	0.10	0.27	0.53
Earnings Beta	-0.43	0.01	-0.07	-0.03	-0.02	0.04	0.10	0.07	0.10	0.57
Bell	-0.69	-0.46	-0.35	-0.19	-0.30	-0.23	-0.01	-0.01	-0.01	0.31
RFD_{CEA}	-0.13	-0.08	-0.08	0.18	0.12	-0.02	0.06	0.06	0.29	0.87
$RFD_{CEA\ SYST}$	-0.06	-0.04	-0.14	-0.12	-0.05	-0.05	0.04	0.06	0.29	0.94
$RFD_{CEA\ IDIO}$	-0.14	-0.18	-0.12	-0.03	-0.02	-0.02	0.12	0.22	0.28	0.86
RFD_{KM}	-0.31	-0.04	-0.13	-0.15	-0.18	0.13	-0.05	0.31	0.39	0.69
RFD_{LM}	-0.37	-0.05	-0.11	-0.15	-0.20	-0.03	0.15	0.27	0.36	0.63
PRD	-0.68	-0.45	-0.30	-0.21	-0.02	-0.09	-0.01	0.09	0.05	0.32

Panel C: Factor-mimicking Portfolio Loadings – Cross Sectional Step Controlling for *MKTRF*, *SMB*, and *HML*

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
MRisk	*Size*	*Market Beta*	*Earnings Beta*	*Bell*	RFD_{CEA}	$RFD_{CEA\ SYS}$	$RFD_{CEA\ IDIO}$	RFD_{KM}	RFD_{LM}	*PRD*
0.54* **	-0.11	0.34	0.03*	0.08	0.01	0.02	0.01	0.02	0.03	0.03

This table presents portfolio-level asset pricing tests. Panel A reports value-weighted and equally weighted average monthly excess returns (in percentages) over the 12-month period following the 10-K filing date, excluding the month of the subsequent filing. Stocks are sorted into quartile portfolios based on *MRisk* and alternative macroeconomic proxies. Panel B presents factor loadings (i.e., betas) from time-series regressions in which monthly excess returns are regressed on factor-mimicking portfolios. These portfolios are constructed by ranking firms into deciles based on *MRisk* and alternative proxies, following a methodology analogous to the construction of the Small Minus Big (SMB) and High Minus Low (HML) factors in Fama and French (1993). The *MRisk*-based factor mimics the return of a long-short strategy that takes a long position in firms with high *MRisk* and a short position in those with low *MRisk*. Panel C reports results from cross-sectional regressions testing whether the betas estimated in Panel B explain the average returns across portfolios—i.e., whether portfolios with greater sensitivity to macroeconomic proxies earn higher average returns. These regressions control for exposures to the market risk premium (MKTRF), SMB, and HML. Coefficients are scaled by a factor of 100 for ease of interpretation. Standard errors are clustered by month-year. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively (two-tailed tests).

TABLE 9
Firm-level Asset Pricing Tests

Independent Variables $_{i,t}$	Dependent Variable: ***Monthly Excess Return*** $_{i,\,t+1}$					
	Fixed-Effects OLS			*Fama-MacBeth*		
	(1)	(2)	(3)	(4)	(5)	(6)
MRisk	**0.13*** **	**0.13*** **	**0.11*** **	**0.01** **	**0.01** **	**0.01** **
	(0.00)	**(0.00)**	**(0.00)**	**(0.03)**	**(0.02)**	**(0.04)**
Size	-0.01***	-0.01***	-0.01***	-0.01**	-0.01**	-0.01
	(0.00)	(0.00)	(0.00)	(0.02)	(0.03)	(0.15)
Market Beta	-0.01	-0.01	-0.01	-0.00	-0.00	-0.00
	(0.23)	(0.19)	(0.60)	(0.67)	(0.56)	(0.29)
Earnings Beta			-0.00			-0.00
			(0.78)			(0.15)
$RFD_{CEA\ SYST}$	0.18	0.22	0.14	-0.13	-0.19	-0.22
	(0.44)	(0.38)	(0.38)	(0.68)	(0.54)	(0.41)
$RFD_{CEA\ IDIO}$	0.10	0.08	0.05	0.15	0.22*	0.54
	(0.32)	(0.39)	(0.98)	(0.11)	(0.08)	(0.19)
RFD_{KM}	0.01	0.01*	0.01	0.00	0.00	0.00
	(0.14)	(0.09)	(0.67)	(0.12)	(0.15)	(0.63)
RFD_{LM}	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	(0.18)	(0.19)	(0.56)	(0.48)	(0.41)	(0.71)
PRD	0.00	0.00	0.00	0.00	0.00	0.00
	(0.30)	(0.32)	(0.37)	(0.12)	(0.16)	(0.19)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Return $_t$	No	Yes	Yes	No	Yes	Yes
Month-Year FE	Yes	Yes	Yes	-	-	-
Firm FE	Yes	Yes	Yes	-	-	-
Observations	846,319	828,276	528,885	846,319	828,276	528,885
Adjusted R^2	0.17	0.17	0.17	0.06	0.07	0.07

This table presents results from regressions of future 12-month monthly excess stock returns on *MRisk*, alternative macroeconomic proxies, and firm-level controls. The dependent variable is the monthly excess return over the risk-free rate during the 12-month period following the 10-K filing date, excluding the month in which the subsequent 10-K is filed. The key independent variable, *MRisk*, is defined as the number of macroeconomic terms per 1,000 words in firm *i*'s 10-K filing for year *t*. Control variables include book-to-market ratio, return on assets (ROA), leverage, special items, number of geographic segments, and idiosyncratic stock returns volatility over the prior 12 months. The variable *Bell* is excluded due to its time-invariant nature. Columns (1)–(3) report OLS estimations with month, year, and firm fixed effects. *p*-values (in parentheses) are based on standard errors clustered by month, year, and firm. Columns (4)–(6) present Fama-MacBeth estimations using Newey-West corrections (with a lag of 3 years or 36 months). Columns (3) and (6) additionally control for *Earnings Beta*, which reduces the sample size to 528,885 firm-month observations. OLS specifications in columns (2)-(3) and (5)-(6) also include the current-month stock return (i.e., Return $_t$) as a control. All continuous variables are winsorized at the 1st and 99th percentiles. All regressions include an intercept (not reported). Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively (two-tailed tests).